

Introduction to Integrating Genomic Data

HUGEN 2073

Genomic Data Visualization and Integration

Jon Chernus

Agenda for second half of HUGEN 2073

- Identify important repositories of information relevant to genetic and genomic research questions and describe the available data contained therein.
- Annotate genes, genetic variation and genomic regions with information from such repositories using local, computational cluster and web-based software interfaces.

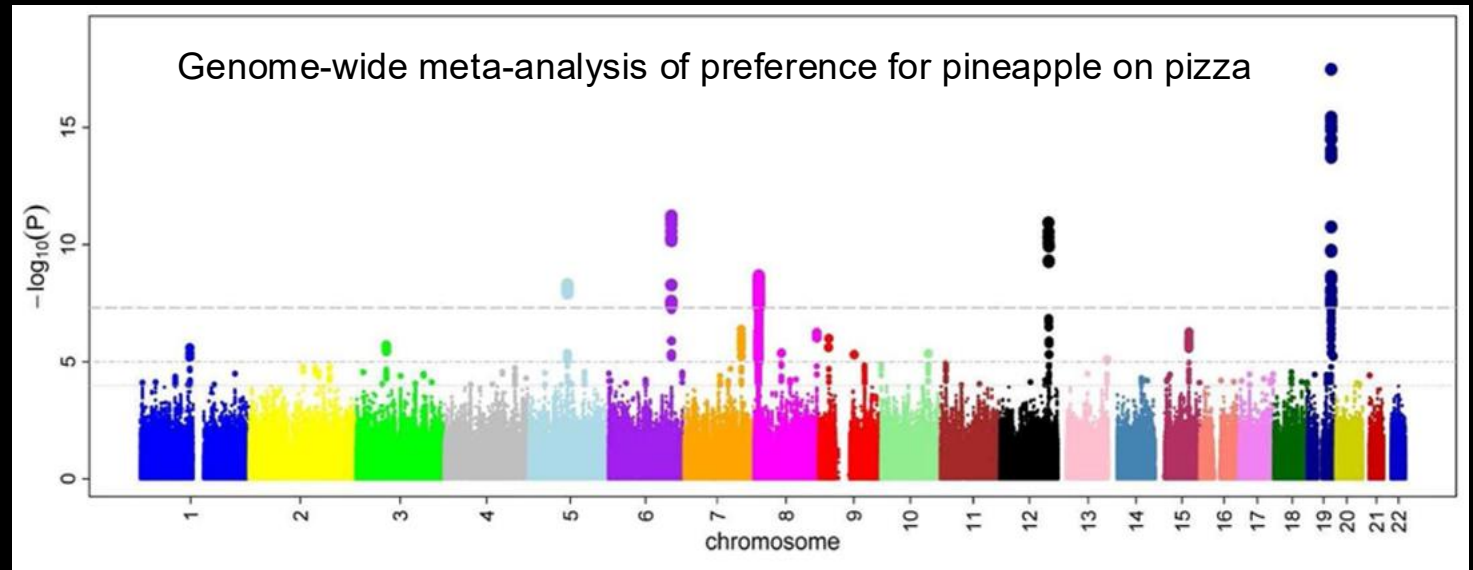
This lecture

- Why should we do annotation/integration?
- Overview of what kind of information is “out there”
- Overview of a few databases and tools
- Two examples of “input” and “output”
- Some limitations

Association results

Everything above the dotted lines is a pineapple-on-pizza gene, right?
Your GWAS is done, and you can publish it now, right?

SNP	P
chr1:135367430	2.49E-01
chr1:111988570	1.74E-02
chr1:112070215	5.41E-01
chr1:112091824	1.04E-01
chr2:102082746	9.47E-01
chr2:102086444	4.71E-01
chr2:102118534	3.59E-01
chr2:102129814	4.93E-01
...	..
...	..
chr22:14567264	9.22E-03



Nope. But why?

What is purpose/goal of a GWAS (not just a list of p-values)?

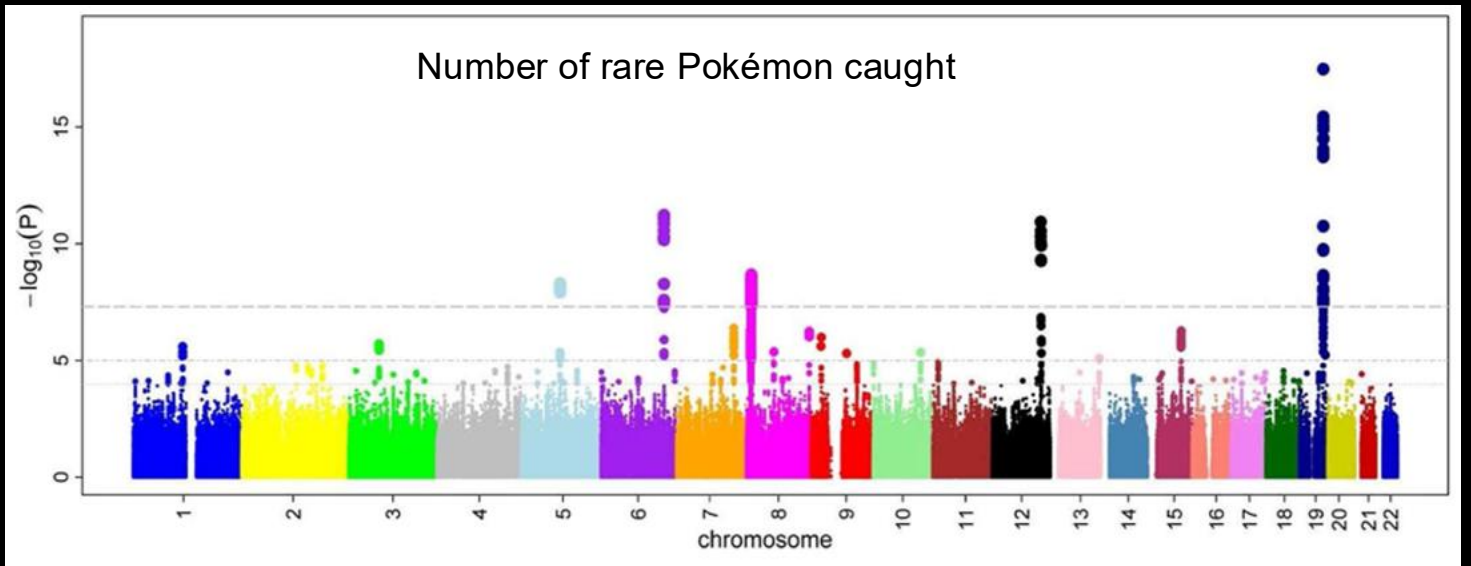
1. GWAS
2. ???*
3. Insight into biology; new hypotheses

*The second half of this class is mostly about Step 2.

Association results

We can just list “the genes”, then, right?

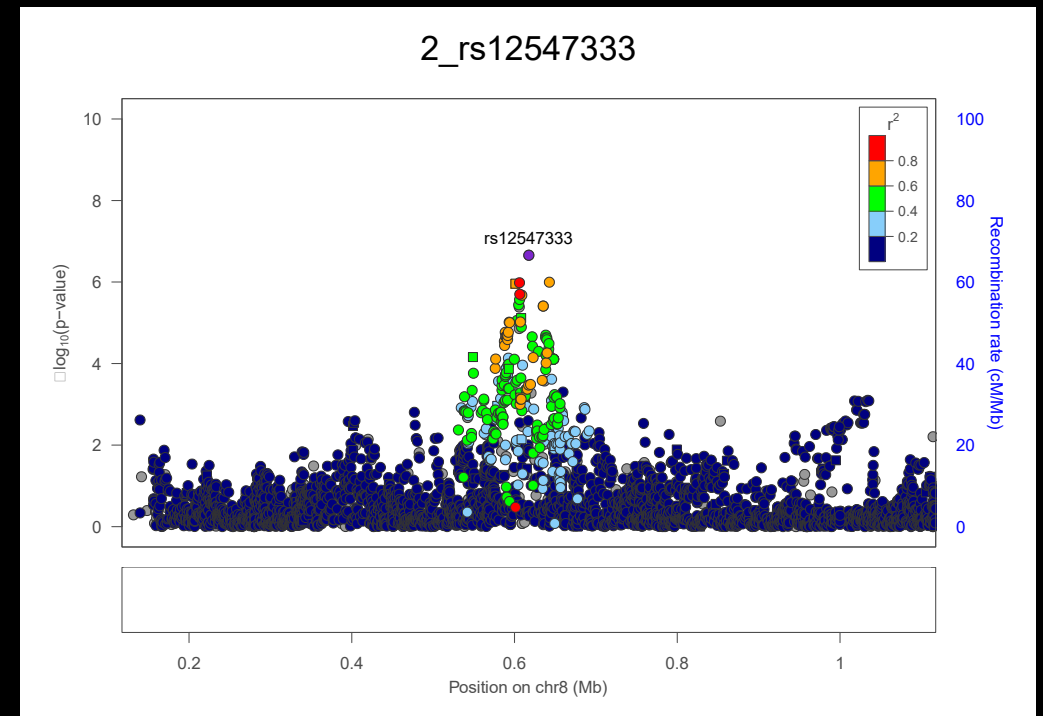
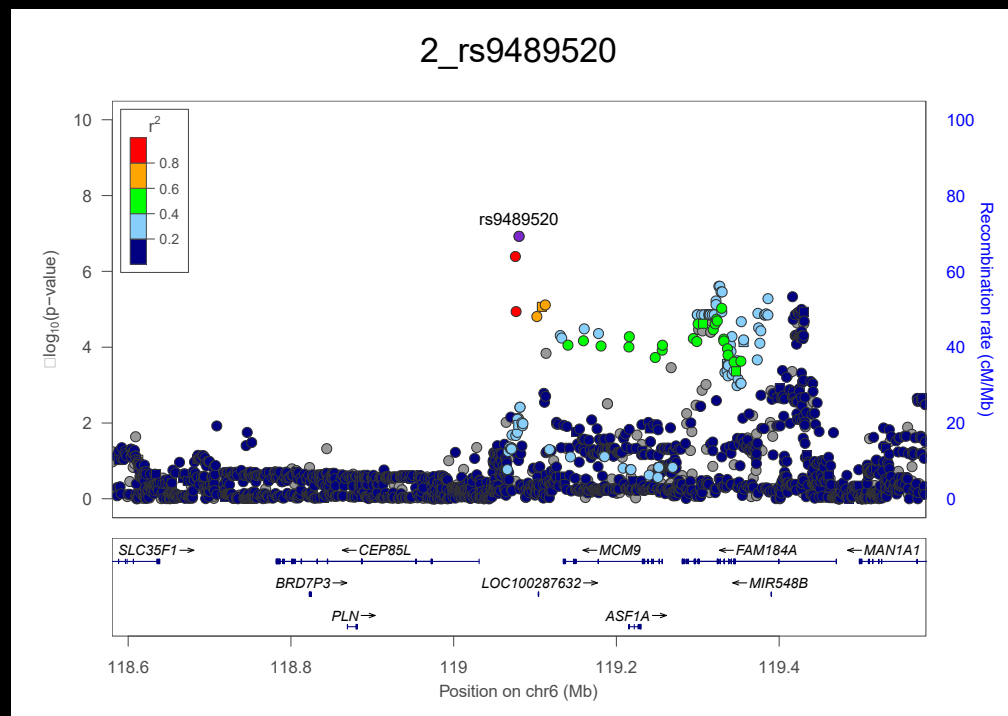
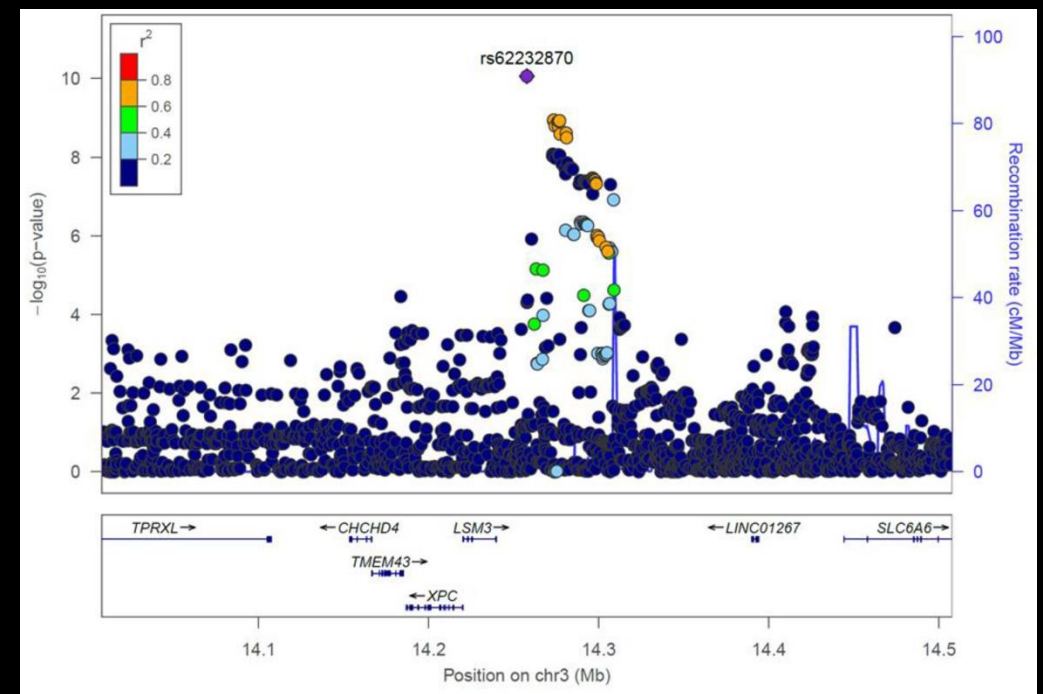
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chr2:102118534	3.59E-01
chr2:102129814	4.93E-01
...	...
...	...
chr22:14567264	9.22E-03



Not exactly. But why?

What does “the” variant do to “the” gene?
And what if...

- the “signal” spans several genes?
- the “signal” is between genes?
- the “signal” isn’t near any genes?



Linkage disequilibrium

But before even thinking about “SNP → gene/function”, we can’t just assume we know “the” SNP.

Typically there will be a number of significant SNPs at an associated locus. And those are only what you *see*.

Is the causal SNP (if there is one) necessarily among the tested SNPs?

Linkage disequilibrium

Question

- What is LD?
- When would you say “SNP1 and SNP2 are in LD”?

Answer

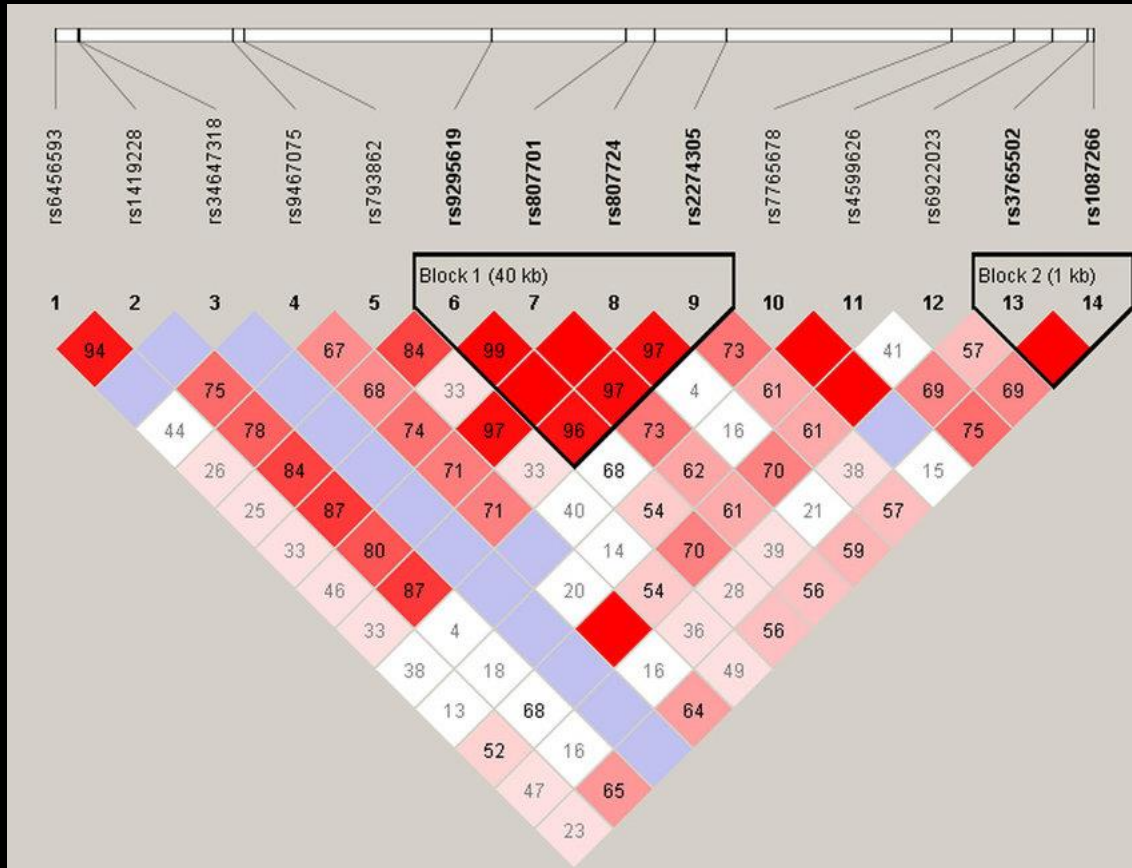
- LD = Association between alleles in a population
- “SNP1 and SNP2 are in LD” means SNP1 and SNP2 are on the same chromosome and that their genotypes are correlated in a population

Implications

- The variants you genotyped help you indirectly “tag” more the genome
- So LD may help you increase chances of detecting an untested “causal” variant
- Strongest-associated variant you saw may not be causal
- LD → more credible set of variants, but lots of variants to sift through

Linkage disequilibrium

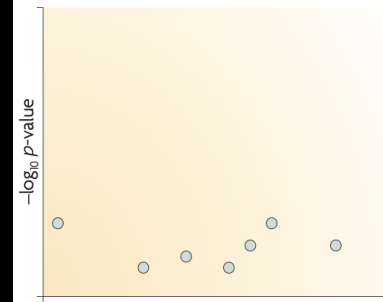
One way of visualizing LD



Recall that LD is why imputation is possible

Box 1 | How genotype imputation works

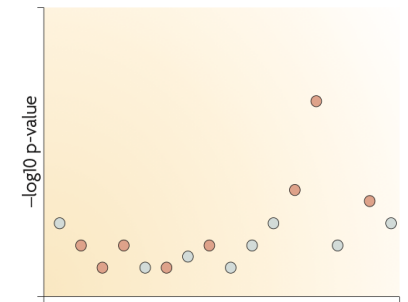
b Testing association at typed SNPs may not lead to a clear signal



d Reference set of haplotypes, for example, HapMap

0	0	0	0	1	1	1	0	0	1	1	1	1	1	1	0
1	1	1	1	1	1	1	0	0	1	0	0	1	1	1	0
1	1	1	1	1	0	1	0	0	1	0	0	0	1	0	1
0	0	1	0	1	1	1	0	0	1	1	1	1	1	1	0
1	1	1	0	1	1	0	0	1	1	1	0	1	1	1	0
0	0	1	0	1	1	1	0	0	1	1	1	1	1	1	0
1	1	1	1	1	0	1	0	1	0	0	0	1	1	0	1
1	1	1	0	0	1	0	0	1	1	1	0	1	1	1	0
0	0	0	0	1	1	1	0	0	1	1	1	1	1	1	0
1	1	1	0	0	1	0	0	1	1	1	0	1	1	1	0

f Testing association at imputed SNPs may boost the signal



a Genotype data with missing data at untyped SNPs (grey question marks)

1	?	?	?	?	1	?	?	1	?	0	2	2	?	?	?	2	?	0
0	?	?	?	?	2	?	?	2	?	0	2	2	?	?	?	2	?	0
1	?	?	?	?	2	?	?	0	2	1	?	?	?	?	?	?	?	0
1	?	?	?	?	2	?	?	1	?	1	2	2	?	?	?	?	?	0
2	?	?	?	?	2	?	?	2	?	1	2	1	?	?	?	?	?	0
1	?	?	?	?	1	?	?	1	?	1	2	2	?	?	?	?	?	1
1	?	?	?	?	2	?	?	2	?	0	2	1	?	?	?	?	?	1
2	?	?	?	?	1	?	?	1	?	1	2	1	?	?	?	?	?	1
1	?	?	?	?	0	?	?	0	?	2	2	2	?	?	?	?	?	0

c Each sample is phased and the haplotypes are modelled as a mosaic of those in the haplotype reference panel

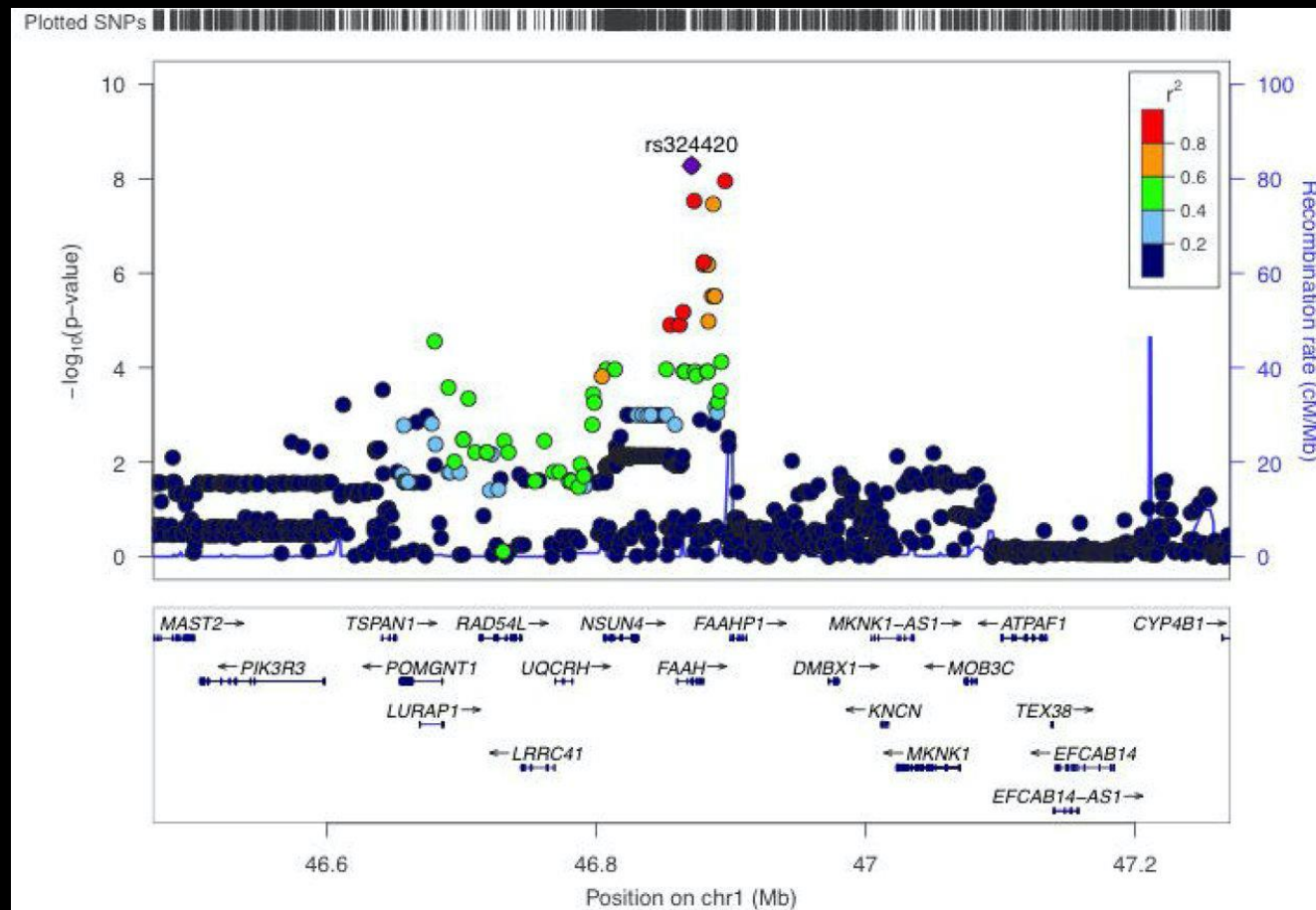
0	?	?	?	?	1	?	?	1	?	0	1	1	?	?	?	1	?	0
1	?	?	?	?	1	?	?	1	?	0	1	1	?	?	?	1	?	0
⋮																		
1	?	?	?	?	1	?	?	1	?	0	1	0	?	?	?	1	?	0
1	?	?	?	?	1	?	?	1	?	1	1	?	?	?	?	1	?	0
⋮																		
1	?	?	?	?	0	?	?	0	?	?	1	1	?	?	?	1	?	0
0	?	?	?	?	0	?	?	0	?	?	1	1	?	?	?	1	?	0

e The reference haplotypes are used to impute alleles into the samples to create imputed genotypes (orange)

1	1	1	1	1	2	1	0	0	2	2	0	2	2	2	2	0
0	0	1	0	2	2	2	0	0	2	2	2	2	2	2	2	0
1	1	1	1	2	2	2	0	0	2	1	1	2	2	2	2	0
1	1	2	0	2	2	1	0	1	2	2	1	2	2	2	2	0
2	2	2	2	2	1	2	0	1	2	2	1	2	2	2	2	0
1	1	1	0	1	2	1	0	1	2	2	1	2	2	2	2	0
1	1	2	1	2	1	2	0	0	2	1	1	1	2	1	1	1
2	2	2	1	1	1	1	0	1	2	1	0	1	2	1	1	1
1	2	2	0	0	2	0	0	2	2	2	1	2	2	2	2	0

Linkage disequilibrium

- Treat the set of SNPs in high LD with the lead variant together as “candidates”
- The “causal” SNP may not be in your data set, but you may be detecting it indirectly



Some considerations so far

- What SNP at this locus “matters”? (Maybe more than one! Or none.)
- What gene at this locus “matters”?
- Even if we somehow know the SNP/gene implicated, how can we gain biological insight?
- Does the set of all associated loci (genes) tell a story?
- How can an association (or set of associations) lead you to biological insights?
- Can we prioritize the association results (i.e., upgrade some variants or genes as candidates)?

What kinds of information will help you answer these questions?

Discussion

- Context 1
You have your association results: a single significant variant
What information do you want about a single variant?
- Context 2
You have your association results: an association “tower” including SNPs of moderate-to-high significance over a 50 kb locus
What would you like to know about a locus?
- Context 3
You have your association results: a set of 12 “towers” like in Context 2, scattered across the genome
What are some questions you could ask about a set of loci?
- Hint: what types of information might be *out there*?

Discussion - responses

Context 1 What do you want to know about a <u>variant</u> ?	Context 2 What do you want to know about a <u>locus</u> ?	Context 3 What do you want to know about a <u>set of loci</u> ?

Can we categorize these types of annotation/annotation techniques?

Two big categories, with some fuzziness

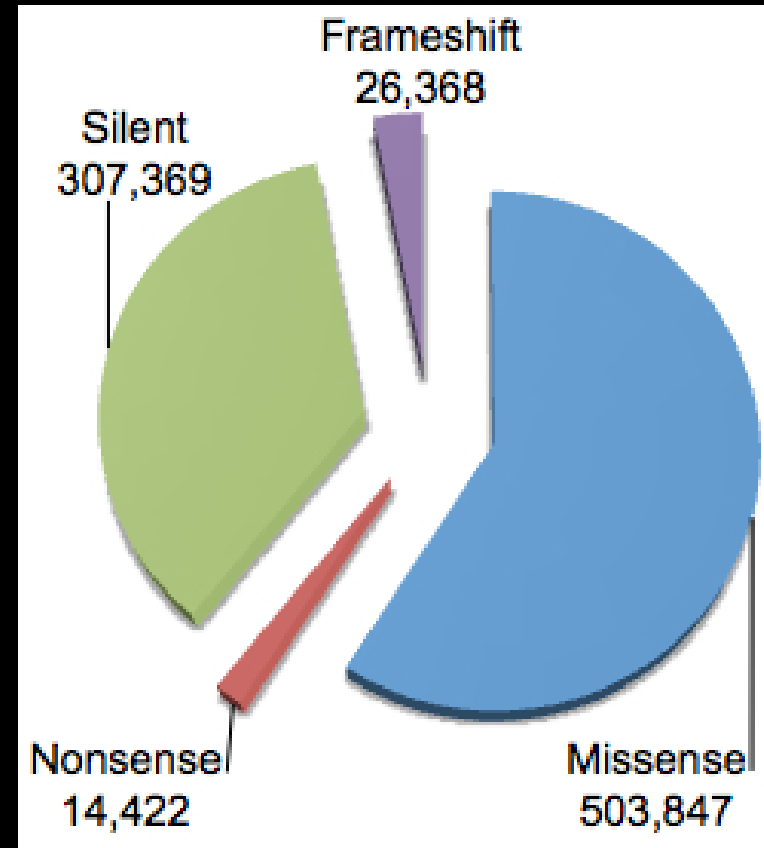
- Positional
- Associational

What kind of positional information is “out there”

- Is the variant of interest in a gene (or in LD with another that is)?
 - List of SNPs in LD (what considerations are there?)
 - Exon locations
 - Intron/splice site locations
 - UTR locations
 - Effects on amino acid sequence (missense, nonsense)
- Other variant-level data
 - E.g., allele frequency in different populations

Coding Variation (Non-Synonymous)

- Initiator codon
- Missense
- Frameshift
- Stop gain
- Stop loss

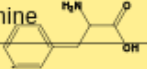
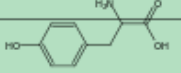
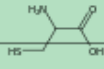
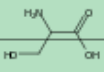
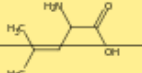
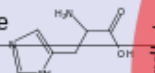
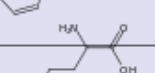
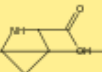
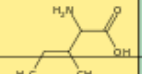
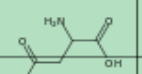
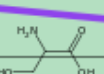
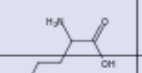
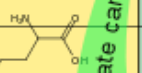
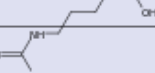
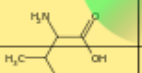
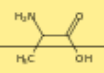
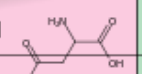
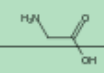


dbSNP 135

Examples of notable Mutations

ΔF508 deletion in cystic fibrosis

1st base 2nd base 3rd base in each row

2nd base			
U	C	A	G
UUU (Phe/F) Phenylalanine 	UCU (Ser/S) Serine	UAU (Tyr/Y) Tyrosine 	UGU (Cys/C) Cysteine 
UUC (Phe/F) Phenylalanine	UCC (Ser/S) Serine 	UAC (Tyr/Y) Tyrosine	UGC (Cys/C) Cysteine
UUA (Leu/L) Leucine	UCA (Ser/S) Serine	UAA Ochre (Stop)	UGA Opal (Stop)
UUG (Leu/L) Leucine	UCG (Ser/S) Serine	UAG Amber (Stop)	UGG (Trp/W) Tryptophan 
CUU (Leu/L) Leucine 	CCU (Pro/P) Proline	CAU (His/H) Histidine 	CGU (Arg/R) Arginine 
CUC (Leu/L) Leucine	CCC (Pro/P) Proline 	CAC (His/H) Histidine	CGC (Arg/R) Arginine
CUA (Leu/L) Leucine	CCA (Pro/P) Proline	CAA (Gln/Q) Glutamine 	CGA (Arg/R) Arginine
CUG (Leu/L) Leucine	CCG (Pro/P) Proline	CAG (Gln/Q) Glutamine 	CGG (Arg/R) Arginine
AUU (Ile/I) Isoleucine 	ACU (Thr/T) Threonine	AAU (Asn/N) Asparagine 	AGU (Ser/S) Serine 
AUC (Ile/I) Isoleucine	ACC (Thr/T) Threonine 	AAC (Asn/N) Asparagine	AGC (Ser/S) Serine
AUA (Ile/I) Isoleucine	ACA (Thr/T) Threonine	AAA (Lys/K) Lysine 	AGA (Arg/R) Arginine 
AUG (Met/M) Methionine 	ACG (Thr/T) Threonine	AAG (Lys/K) Lysine	AGG (Arg/R) Arginine 
GUU (Val/V) Valine 	GCU (Ala/A) Alanine 	GAU (Asp/D) Aspartic acid 	GGU (Gly/G) Glycine 
GUC (Val/V) Valine	GCC (Ala/A) Alanine	GAC (Asp/D) Aspartic acid	GGC (Gly/G) Glycine
GUA (Val/V) Valine	GCA (Ala/A) Alanine	GAA (Glu/E) Glutamic acid 	GGA (Gly/G) Glycine
GUG (Val/V) Valine	GCG (Ala/A) Alanine	GAG (Glu/E) Glutamic acid	GGG (Gly/G) Glycine

Selection of notable mutations, ordered in a standard table of the genetic code of amino acids.

Clinically important missense mutations generally change the properties of the coded amino acid residue between being basic, acidic, polar or nonpolar, while nonsense mutations result in a stop codon.

Amino acids

- Basic
- Acidic
- Polar
- Nonpolar (hydrophobic)

Fragile X Syndrome

Polyglutamine (PolyQ) Diseases

- Huntington's disease
- Spinocerebellar ataxia (SCA) (most types)
- Spinobulbar muscular atrophy (Kennedy disease)
- Dentatorubral-pallidoluysian atrophy

Mutation type

- Trinucleotide repeat
- Deletion
- Missense
- Nonsense

Sickle-cell disease

Friedreich's ataxia

β-Thalassemia

McArdle's disease

β-Thalassemia

Prostate cancer

Colorectal cancer

Examples of databases and tools for annotating single variants

- dbSNP
- gnomAD
- genome browser(s)
- VEP (Variant Effect Predictor)
- ANNOVAR

rs12345

Organism

Homo sapiens

Clinical Significance

Not Reported in ClinVar

Position

chr22:25459492 (GRCh38.p12)

Gene :

CRYBB2P1 : Non Coding Transcript Variant

Alleles

G>A / G>C

Gene Consequence

0 citations

Variation Type

SNV Single Nucleotide Variation

Publications

See rs on genome

Frequency

A=0.049153 (6172/125568, TOPMED)
A=0.04517 (1412/31258, GnomAD)
A=0.0229 (193/8412, ALFA Project) (+ 10 more)

Genomic View

Variant Details

Genomic Placements

Clinical Significance

Frequency

HGVS

Submissions

History

Publications

Flanks

Sequence name	Change
GRCh37.p13 chr 22	NC_000022.10:g.25855459G>A
GRCh37.p13 chr 22	NC_000022.10:g.25855459G>C
GRCh38.p12 chr 22	NC_000022.11:g.25459492G>A
GRCh38.p12 chr 22	NC_000022.11:g.25459492G>C

Gene: **CRYBB2P1**, crystallin beta B2 pseudogene 1 (plus strand)

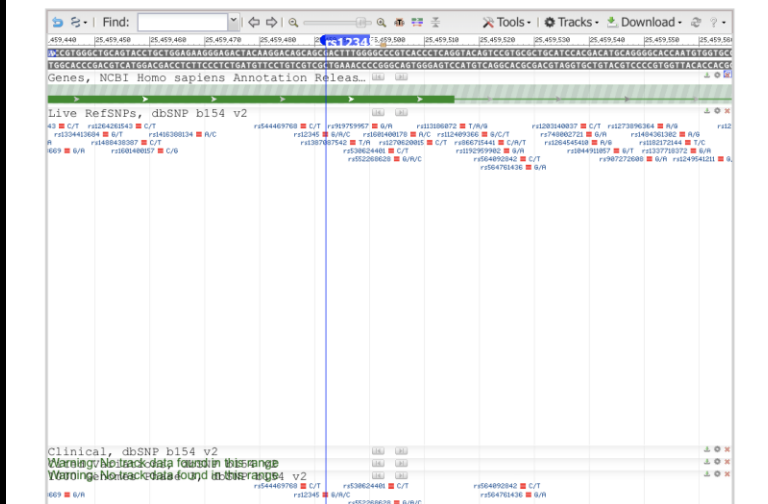
Molecule type	Change	Amino acid[Codon]	SO Term
CRYBB2P1 transcript variant 1	NR_033733.1:n.666G>A	N/A	Non Coding Transcript Variant
CRYBB2P1 transcript variant 1	NR_033733.1:n.666G>C	N/A	Non Coding Transcript Variant
CRYBB2P1 transcript variant 2	NR_033734.1:n.547G>A	N/A	Non Coding Transcript Variant
CRYBB2P1 transcript variant 2	NR_033734.1:n.547G>C	N/A	Non Coding Transcript Variant

Genomic regions, transcripts, and products

Choose placement

GRCh38.p12 [NC_000022.11]

See rs12345 in Variation Viewer



← dbSNP
<https://www.ncbi.nlm.nih.gov/snp/>

gnomAD →
<https://gnomad.broadinstitute.org>

Single nucleotide variant: 7-25049919-T-G (GRCh37)

Dataset

gnomAD v2.1.1

Filter

No variant

Exomes

Genomes

Total

Allele Count

31324

609

609

Allele Number

0.01944

31324

0.01944

Allele Frequency

0.02430

0.01944

0.01944

Popmax Filtering AF (95% confidence)

8

8

8

External Resources

- dbSNP (rs156)
- UCSC

Report

- Report this variant
- Request additional information

Variant Effect Predictor

This variant falls on 0 transcripts in 0 genes.

Note The gene symbols shown below are provided by VEP and may differ from the symbol shown on gene pages.

Population Frequencies

Population	Allele Count	Allele Number	Number of Homozygotes	Allele Frequency
European (Finnish)	133	3436	0	0.03871
Ashkenazi Jewish	11	290	0	0.03793
European (non-Finnish)	407	15410	8	0.02641
Other	27	1084	0	0.02491
Latino/Admixed American	3	840	0	0.003571
African/African-American	28	8704	0	0.003217
East Asian	0	1560	0	0.000
South Asian	0	0	0	0.000
XX	291	13882	1	0.02096
XY	318	17442	7	0.01823
Total	609	31324	8	0.01944

Include: Exomes Genomes

Age Distribution

Sequencing types: Genome

Include heterozygous variant carriers

Include homozygous variant carriers

Compare to all individuals

Genotype Quality Metrics

Sequencing types: Genome

Compare to all individuals

Site Quality Metrics

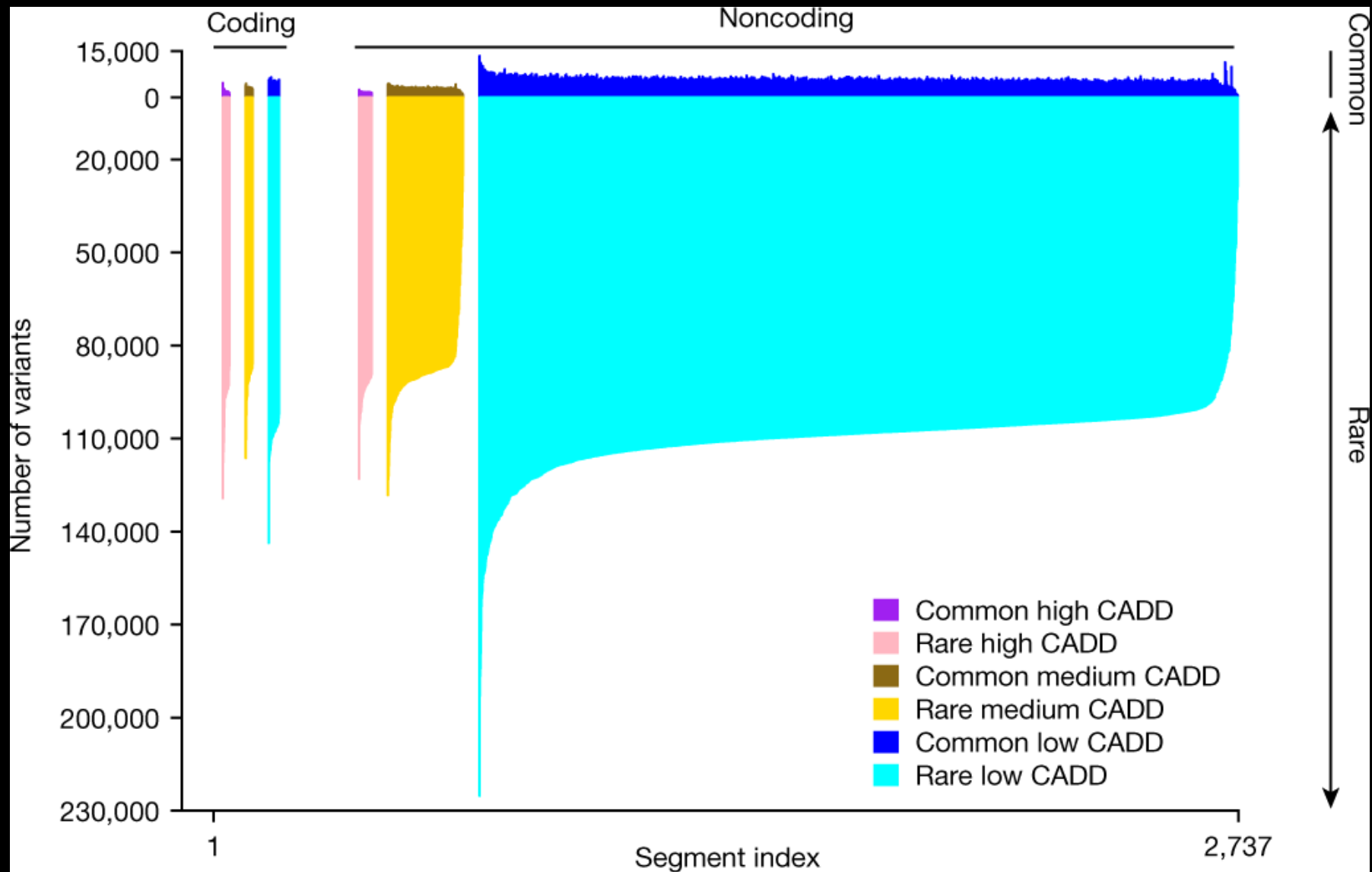
Sequencing types: Genome

Value: 2.852e+5 (genome samples)

This is the site quality distribution for all genome variants with 0.01 <= AF < 0.02.

Note: These are site-level quality metrics, they may be unpredictable for multi-allelic sites.

TOPMED



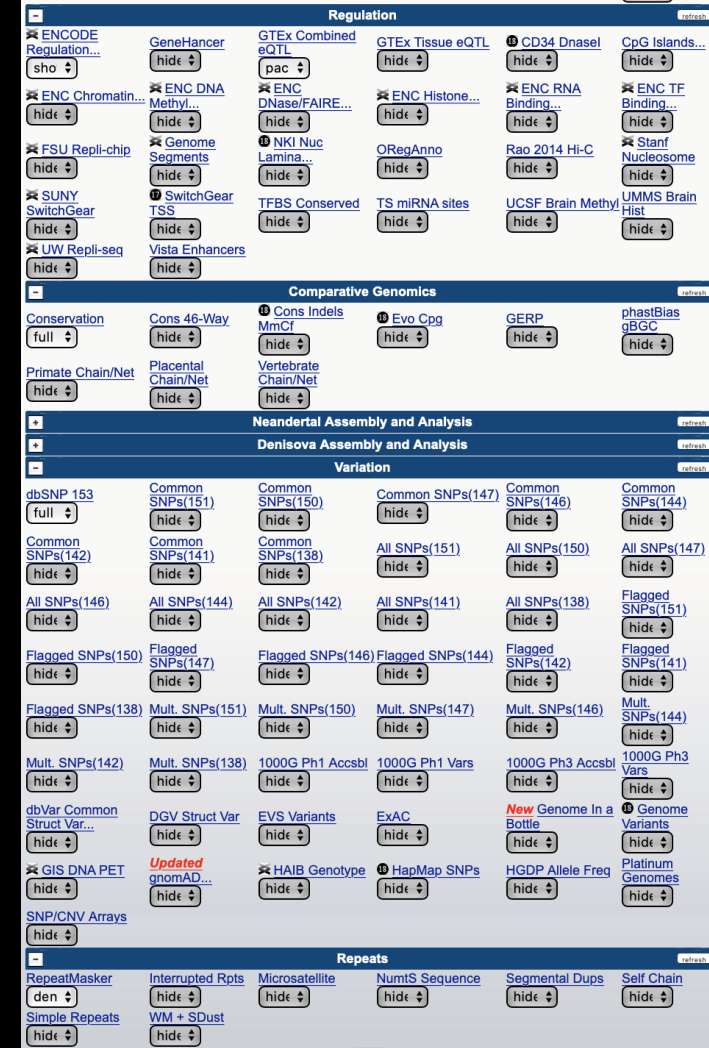
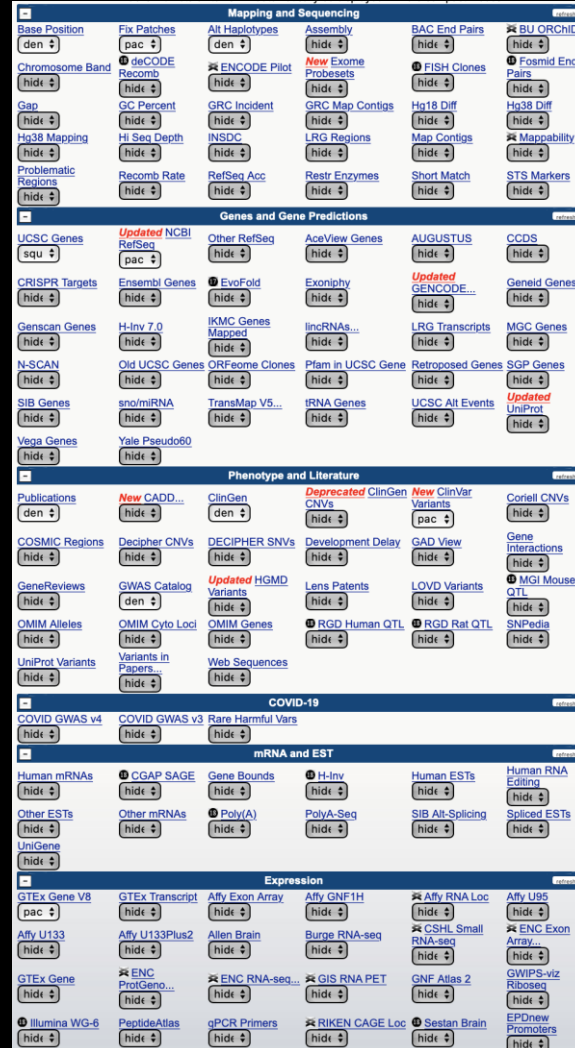
UCSC Genome Browser

You will learn to extract underlying data and to add your own tracks.



chr9:123,873,182-125,883,181 2,010,000 bp.

[Video on the Coronavirus Browser](#)



More variant-level information

- Known phenotype associations
 - OMIM (Online Mendelian Inheritance in Man)
 - GWAS Catalog
 - PheWeb
 - More
- LD
 - TOPMED
 - 1000 Genomes Project
 - More
- Other, including predicted effects
 - SIFT (conservation score)
 - PolyPhen (non-synonymous SNP deleteriousness score)
 - CADD (a composite deleteriousness score)

*604105

Table of Contents

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Gene–Phenotype Relationships

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Contributors

Creation Date

Edit History

* 604105

SYNAPTONEMAL COMPLEX PROTEIN 2; **SYCP2**

Alternative titles; symbols

SCP2

HGNC Approved Gene Symbol: **SYCP2**

Cytogenetic location: **20q13.33**

Genomic coordinates (GRCh38): **20:59,863,276-59,933,699** (from NCBI)

Gene-Phenotype Relationships

Location	Phenotype	Phenotype MIM number	Inheritance	Phenotype mapping key
20q13.33	Spermatogenic failure 1	258150	AR	3

PheneGene Graphics - ?

TEXT

▼ Cloning and Expression

During meiotic prophase, chromosomes are arranged along proteinaceous axes called axial elements. In rat, the major protein components of axial elements are proteins of 30, 33, and 190 kD. The 30- and 33-kD proteins appear to be products of a single gene, Scp3 (SYCP3; **604759**). [Offenberg et al. \(1998\)](#) isolated rat testis cDNAs encoding the 190-kD protein, which they designated Scp2. Scp2 is a basic protein, with a pI of 8. It contains 2 clusters of S/T-P motifs, which are common in DNA-binding proteins, and a C-terminal coiled-coil region. Scp2 shows some similarity at the amino acid sequence and secondary structural levels to the *S. cerevisiae* Red1 protein, which is involved in meiotic recombination and the assembly of axial elements of synaptonemal complexes. Northern blot analysis showed that Scp2 was expressed exclusively in rat testis. ImmunocytoLOGY localized Scp2 specifically to the synaptonemal complex in the nuclei of rat testis meiotic prophase nuclei. ?

By screening a human testis library with a partial rat Scp2 cDNA, [Schalk et al. \(1999\)](#) isolated cDNAs encoding human SCP2. The predicted 1,530-amino acid human protein shares 63% amino acid identity with rat Scp2. Like rat Scp2, human SCP2 contains S/T-P motifs, 2 nuclear targeting signals, and a C-terminal coiled-coil region. ?

▼ Mapping

By FISH, [Schalk et al. \(1999\)](#) mapped the human SCP2 gene to chromosome 20q13.33. ?

▼ Gene Function

[Offenberg et al. \(1998\)](#) found that recombinant rat Scp2 bound DNA in Southwestern blot experiments. They speculated that Scp2 is a DNA-binding protein involved in the structural organization of meiotic prophase chromosomes. ?

Using protein interaction assays, [Yang et al. \(2006\)](#) showed that mouse Sycp2 interacted directly with Sycp3 in vitro and in vivo, and that the interaction required the coiled-coil domain of **Sycp2**. Western blot analysis of chromatin extracts of mouse testis detected **Sycp2**, as well as Sycp1 (**602162**), Sycp3, and histone H3 (see **602810**). ?

▼ External Links

▶ Genome

▶ DNA

▶ Protein

▶ Gene Info

▶ Clinical Resources

▼ Variation

1000 Genome

ClinVar

gnomAD

GWAS

Central

HGMD

HGVS

NHLBI EVS

PharmGKB

▶ Animal Models

▶ Cellular Pathways

*604105

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▼ Animal Model

[Yang et al. \(2006\)](#) developed a line of mice expressing a mutant **Sycp2** in which exon 38 was fused in-frame to exon 44. The mutant protein was expressed, although it was truncated and lacked the C-terminal coiled-coil domain. Homozygous mutant mice were obtained at a normal mendelian ratio and appeared normal, but males were sterile, and females were subfertile with sharply reduced litter size. Heterozygous **Sycp2** mutant males and females showed normal fertility. Homozygous **Sycp2** mutant spermatocytes exhibited complete meiotic arrest, failed to differentiate into pachytene spermatocytes, and underwent apoptosis in type II seminiferous tubules due to failure in the formation of axial elements and chromosomal synapsis. Homozygous **Sycp2** mutant fetal oocytes showed homologous chromosome alignment, but full chromosome synapsis was interrupted by prominent axial gaps in the synaptonemal complex. The mutant **Sycp2** protein localized to axial chromosomal cores in both spermatocytes and fetal oocytes, but Sycp3 did not and formed nuclear aggregates. Furthermore, mutant **Sycp2** only loosely bound chromatin. [Yang et al. \(2006\)](#) concluded that **SYCP2** is a primary determinant of axial elements and lateral elements and is required for incorporation of SYCP3 into synaptonemal complexes. ?

▼ Cytogenetics

In a 28-year-old man (DGAP230) with a 2-year history of infertility and severe oligozoospermia (see **258150**), [Schilit et al. \(2020\)](#) identified the apparently balanced reciprocal translocation 46,XY,t(20;22)(q13.3;q11.2). Breakpoint analysis yielded 20 candidate genes that were disrupted or potentially dysregulated, and expression analysis by qPCR revealed that **SYCP2** expression in the patient's lymphoblastoid cell line (LCL) was increased more than 20-fold compared to control LCLs. Misexpression of **SYCP2** appeared to derive exclusively from the der(20) allele, and further analysis of the der(20) chromosome supported an enhancer adoption model, in which an active enhancer residing in the segment of chromosome 22 translocated to the der(20) may enter a newly formed chromatin contact encompassing the cis-residing **SYCP2** allele, resulting in illegitimate overexpression. Overexpression of the yeast functional homolog of **SYCP2**, RED1, resulted in aberrant assembly of the synaptonemal complex in budding yeast meiotic nuclei. The authors suggested that an abnormal abundance of **SYCP2** axial element protein disrupted homologous chromosome synapsis in patient DGAP230, resulting in severe oligozoospermia due to asynapsis-triggered cell death in spermatocytes. ?

▼ Molecular Genetics

[Schilit et al. \(2020\)](#) examined exome sequencing data from a male infertility cohort and identified 3 men with oligo- or azoospermia (spermatogenic failure-1, SPGF1; **258150**) who had heterozygous truncating variants in the **SYCP2** gene (**604105.0001-604105.0003**) that were not found in the gnomAD or TOPMed databases. ?

▼ ALLELIC VARIANTS (3 Selected Examples):

Table View

ClinVar

.0001 SPERMATOGENIC FAILURE 1

SYCP2, 5-BP DEL, NT3067

rs753462162 | RCV001005040...

In a 39-year-old man (M1401) with infertility and azoospermia, who had meiotic arrest at the pachytene spermatocyte stage (SPGF1; **258150**), [Schilit et al. \(2020\)](#) identified heterozygosity for a 5-bp deletion (c.3067_3071del, NM_014258.3) in exon 33 of the **SYCP2** gene, causing a frameshift predicted to result in a premature termination codon (Lys1023LeufsTer2). The mutation was not found in the gnomAD or TOPMed databases. ?

.0002 SPERMATOGENIC FAILURE 1

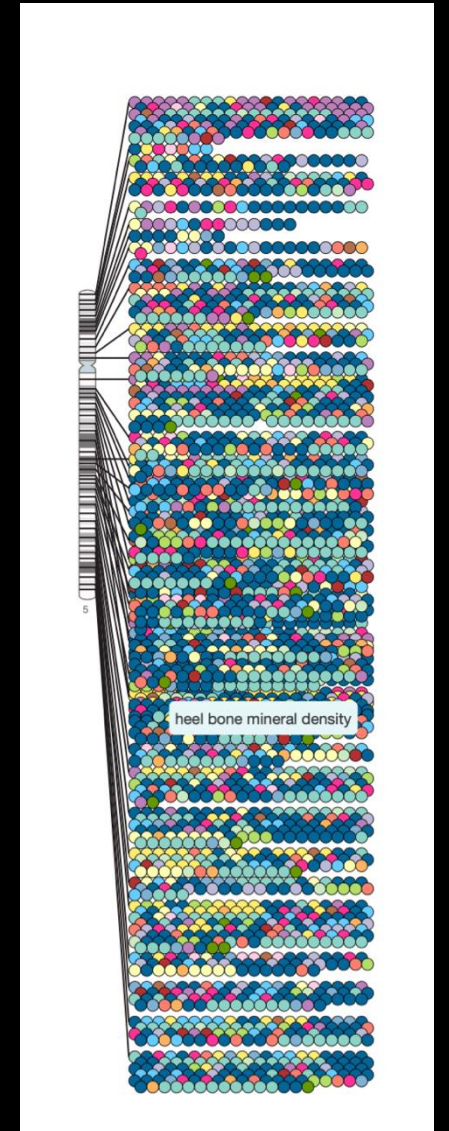
SYCP2, 5-BP DEL, NT2793

rs1600840291 | RCV000855716...

In a 27-year-old infertile man (M1581) with cryptozoospermia (SPGF1; **258150**), [Schilit et al. \(2020\)](#) identified heterozygosity for a 5-bp deletion (c.2793_2797del, NM_014258.3) in exon 31 of the **SYCP2** gene, causing a frameshift predicted to result in a premature termination codon (Lys932SerfsTer3). The mutation, which was inherited from his unaffected mother, was not found in the gnomAD or TOPMed databases. ?

GWAS Catalog

<https://www.ebi.ac.uk/gwas/>



Variant Effect Predictor (VEP)

ensembl.org/Tools/VEP

The screenshot shows the Ensembl VEP web interface. The top navigation bar includes links for BLAST/BLAT, VEP, Tools, BioMart, Downloads, and More. A search bar for species is present. The left sidebar lists various web tools, with 'Variant Effect Predictor' highlighted. The main content area is titled 'Variant Effect Predictor' and contains a 'New job' button and a 'Clear form' button. The 'Species' dropdown is set to 'Human (Homo sapiens)'. Below it, the 'Assembly' is specified as 'GRCh38.p13'. The 'Name for this job (optional):' field is empty. The 'Input data:' section has a text area for pasting data and a file upload button. The 'Transcript database to use:' section has radio buttons for 'Ensembl/GENCODE transcripts' (selected), 'Ensembl/GENCODE basic transcripts', 'RefSeq transcripts', and 'Ensembl/GENCODE and RefSeq transcripts'. The 'Additional configurations:' section has a dropdown for 'Identifiers' and a text input for 'Additional identifiers for genes, transcripts and variants'. A footer notice states: 'This website requires cookies, and the limited processing of your personal data in order to function. By using the site you are agreeing to this as outlined in our Privacy Policy and Terms of Use. I Agree'.

The screenshot shows the 'Results preview' page of the Ensembl VEP. It includes a 'Navigation (per variant)' section with a 'New job' button. The 'Page' is 1 of 1, and 'Show: 1 All variants'. The 'Filters' section shows 'Uploaded variant' is defined. The 'Download' section has links for 'VCF VEP TXT' and 'BioMart: Variants & Genes'. A 'Show/hide columns (12 hidden)' button is present. The table below shows the results of the VEP analysis.

Uploaded variant	Location	Allele	Consequence	Symbol	Gene	Feature type	Feature
AGRN:c.1057C>T	1:1041582-1041582	T	stop_gained	AGRN	ENSG00000188157	Transcript	ENST
AGRN:c.1057C>T	1:1041582-1041582	T	downstream_gene_variant	AGRN	ENSG00000188157	Transcript	ENST
AGRN:c.1057C>T	1:1041582-1041582	T	upstream_gene_variant	AGRN	ENSG00000188157	Transcript	ENST
AGRN:c.1057C>T	1:1041582-1041582	T	stop_gained	AGRN	ENSG00000188157	Transcript	ENST
AGRN:c.1057C>T	1:1041582-1041582	T	stop_gained	AGRN	ENSG00000188157	Transcript	ENST
AGRN:c.1057C>T	1:1041582-1041582	T	stop_gained	AGRN	ENSG00000188157	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	missense_variant	APOE	ENSG00000130203	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	missense_variant	APOE	ENSG00000130203	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	missense_variant	APOE	ENSG00000130203	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	missense_variant	APOE	ENSG00000130203	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	downstream_gene_variant	APOE	ENSG00000130203	Transcript	ENST
APOE:c.526C>T	19:44908822-44908822	T	upstream_gene_variant	AC011481.3	ENSG00000280087	Transcript	ENST

Page: 1 of 1 | Show: 1 All variants

VEP

Not Secure — useast.ensembl.org

Results preview

Navigation (per variant)

Page: 1 of 1 | Show: 1 All variants

Filters

Uploaded variant is defined Add

Download

All: [VCF](#) [VEP](#) [TXT](#)

BioMart: [Variants](#) [Genes](#)

Show/hide columns (12 hidden)

Feature	Biotype	Exon	cDNA position	CDS position	Protein position	Amino acids	Codons	Existing variant	Feature strand
ST00000379370.7	protein_coding	6/36	1110	1057	353	Q/*	CAG/TAG	rs587777299 , CM126385	1
ST00000469403.1	retained_intron	-	-	-	-	-	-	rs587777299 , CM126385	1
ST00000479707.1	retained_intron	-	-	-	-	-	-	rs587777299 , CM126385	1
ST00000620552.4	protein_coding	6/39	1107	643	215	Q/*	CAG/TAG	rs587777299 , CM126385	1
ST00000651234.1	protein_coding	5/38	1192	742	248	Q/*	CAG/TAG	rs587777299 , CM126385	1
ST00000652369.1	protein_coding	5/35	1192	742	248	Q/*	CAG/TAG	rs587777299 , CM126385	1
ST00000252486.9	protein_coding	4/4	595	526	176	R/C	CGC/TGC	rs7412 , CM860003	1
ST00000425718.1	protein_coding	3/3	791	526	176	R/C	CGC/TGC	rs7412 , CM860003	1
ST00000434152.5	protein_coding	4/4	661	604	202	R/C	CGC/TGC	rs7412 , CM860003	1
ST00000446996.5	protein_coding	4/4	615	526	176	R/C	CGC/TGC	rs7412 , CM860003	1
ST00000485628.2	retained_intron	-	-	-	-	-	-	rs7412 , CM860003	1
ST00000623895.1	TEC	-	-	-	-	-	-	rs7412 , CM860003	1

Page: 1 of 1 | Show: 1 All variants

VEP

Not Secure — useast.ensembl.org

Navigation (per variant)

New job

Page: 1 of 1 | Show: 1 All variants

Filters

Uploaded variant is defined Add

Download

All: [VCF](#) [VEP](#) [TXT](#)

BioMart: [Variants](#) [Genes](#)

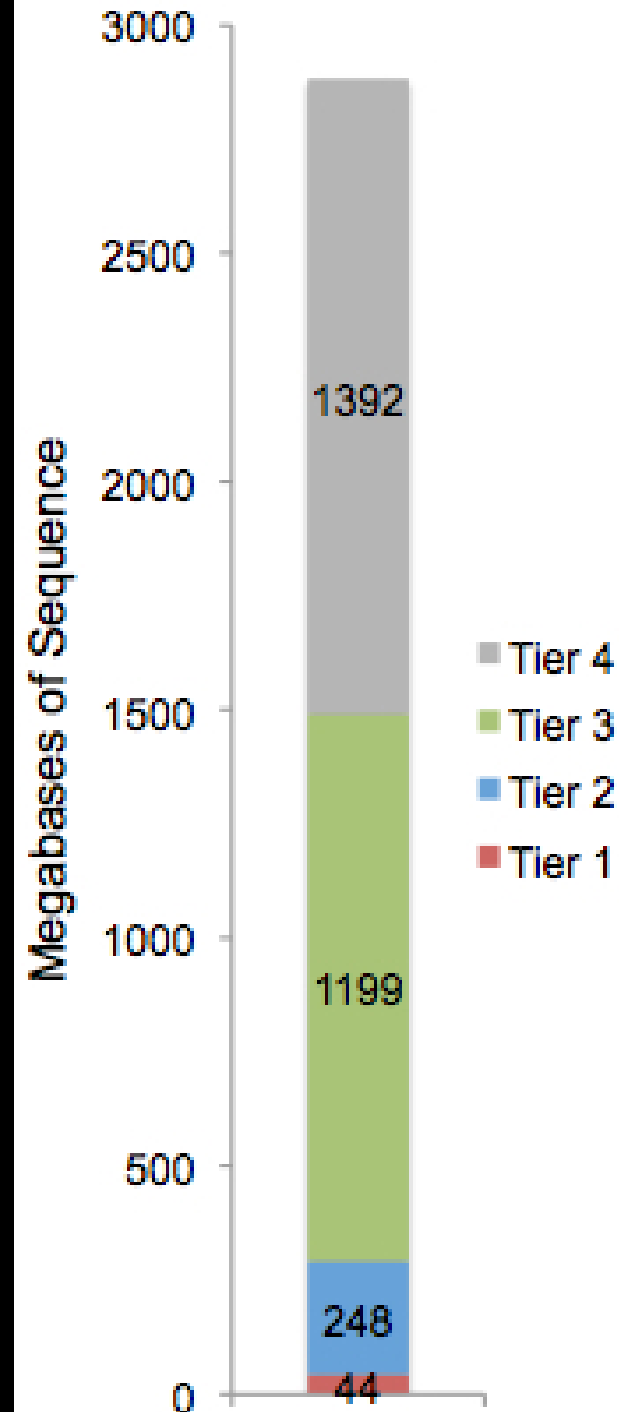
Show/hide columns (12 hidden)											
ns	Existing variant	Feature strand	MANE	Transcript support level	APPRIS	SIFT	PolyPhen	AF	Clinical significance	Pu	
TAG	rs587777299, CM126385	1	NM_198576.4	1	P1	-	-	-	pathogenic	22	
	rs587777299, CM126385	1	-	3	-	-	-	-	pathogenic	22	
	rs587777299, CM126385	1	-	3	-	-	-	-	pathogenic	22	
TAG	rs587777299, CM126385	1	-	5	-	-	-	-	pathogenic	22	
TAG	rs587777299, CM126385	1	-	-	-	-	-	-	pathogenic	22	
TAG	rs587777299, CM126385	1	-	-	-	-	-	-	pathogenic	22	
TGC	rs7412, CM860003	1	NM_000041.4	1	P1	0	1	0.0751	drug_response, other, pathogenic	11	Pu ID:
TGC	rs7412, CM860003	1	-	1	-	0	1	0.0751	drug_response, other, pathogenic	11	Pu ID:
TGC	rs7412, CM860003	1	-	2	-	0	0.999	0.0751	drug_response, other, pathogenic	11	Pu ID:
TGC	rs7412, CM860003	1	-	2	-	0	1	0.0751	drug_response, other, pathogenic	11	Pu ID:
	rs7412, CM860003	1	-	1	-	-	-	0.0751	drug_response, other, pathogenic	11	Pu ID:
	rs7412, CM860003	1	-	-	-	-	-	0.0751	drug_response, other, pathogenic	11	Pu ID:

Page: 1 of 1 | Show: 1 All variants

New job

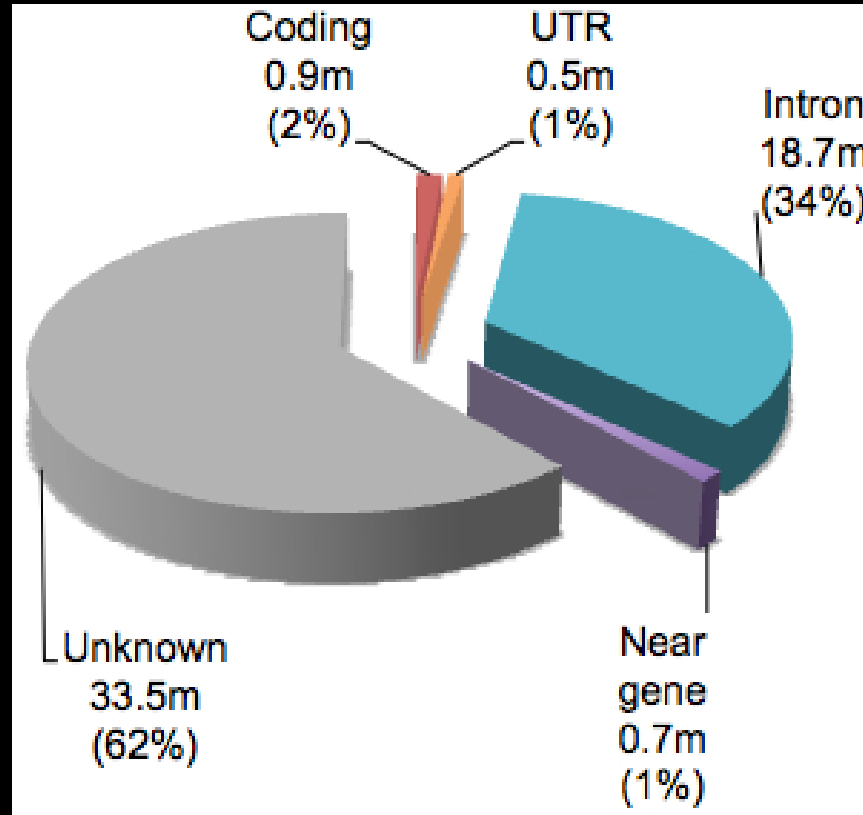
Distribution of Sequence

4. Repetitive regions of the genome
3. Non-coding, non-conserved, unique regions of the genome
2. Evolutionarily conserved or putative regulatory sequences
1. Coding sequences, including exons, splice sites, and non-coding RNA genes

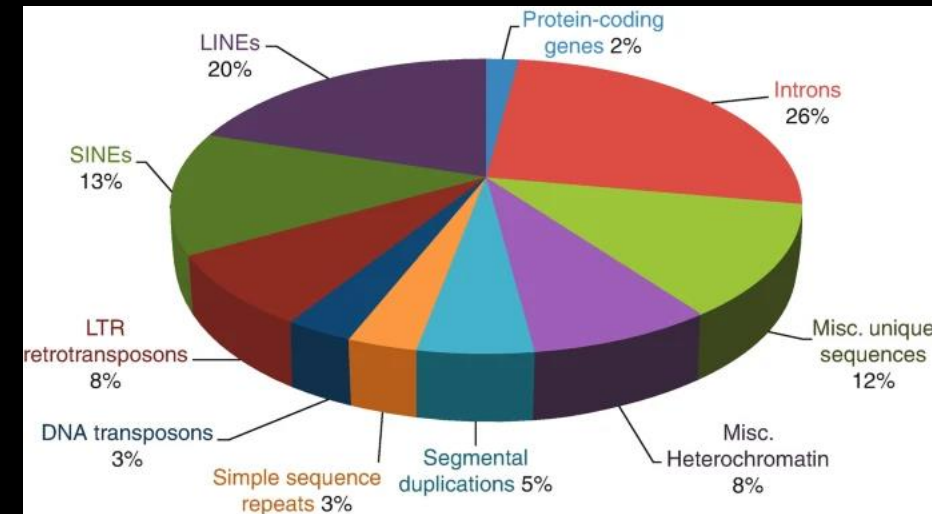


Classes of Functional Variation

- Genic
 - Coding
 - Synonymous
 - Non-synonymous
 - Missense
 - Nonsense
 - Non-Coding
- Intergenic



dbSNP 135*



Entire genome*

*I am really sorry about these plots. It's surprisingly hard to find non-3D graphs of genome composition.

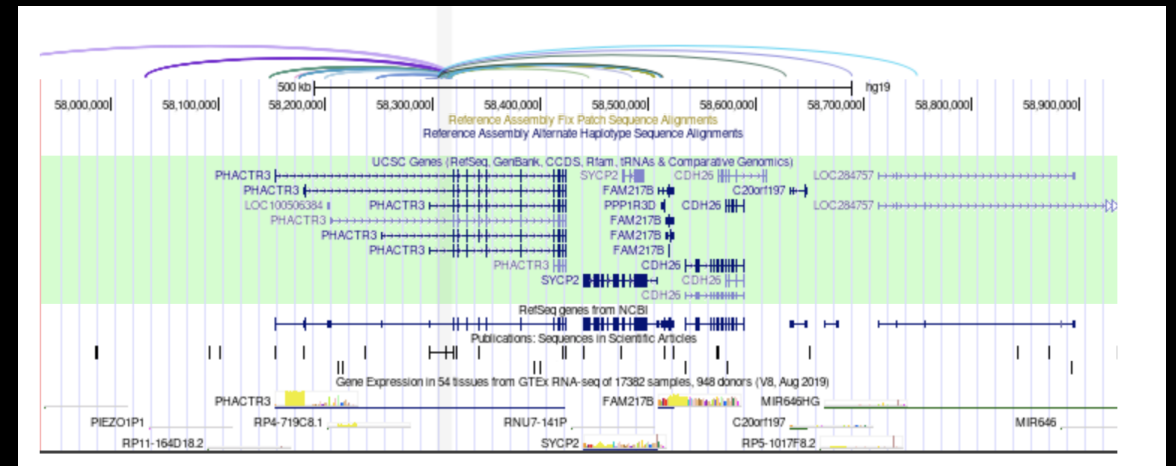
What other positional information is out there?

Most associations won't be situated squarely within one gene.
There may be no gene nearby.

What other kinds of functional elements have been mapped in the genome?

What other “positional” information is out there?

- Is your SNP/variant in a region with any of the following annotations?
 - 5' or 3' UTR
 - Regulatory elements (enhancers, e.g.)
 - Methylation
 - Acetylation
 - Other chromatin state info
 - DNase sensitivity
 - Motifs and conservation
 - eQTLs
 - 3-D interaction, topological domains



Enhancer Variation

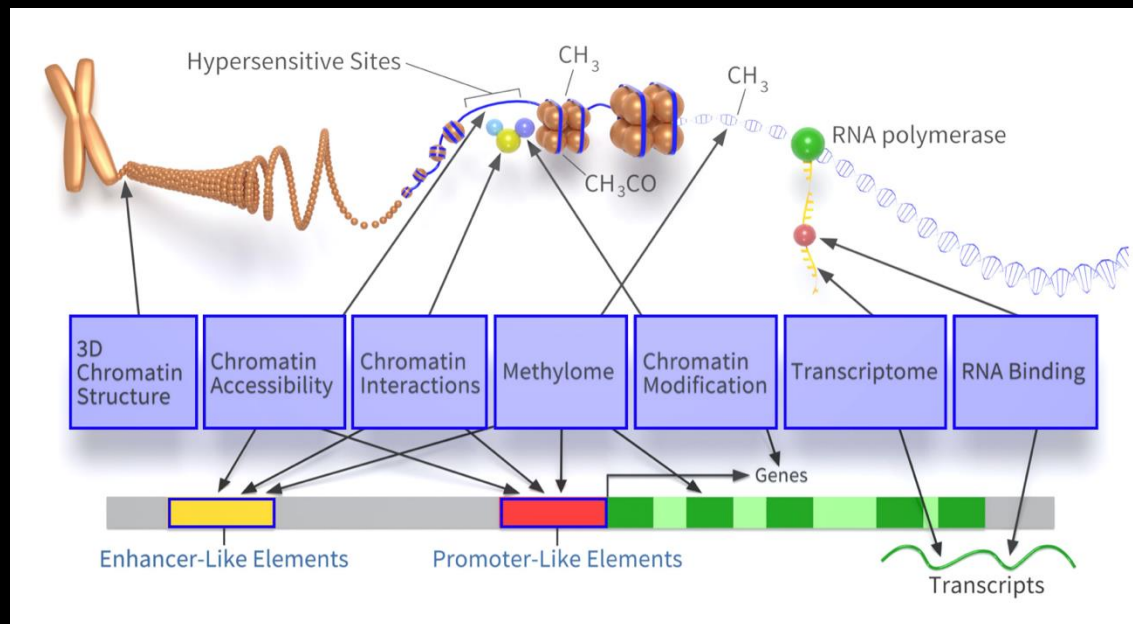
- Enhancers are distal regulatory elements that often lie 10 kbp–100 kpb (sometimes up to 1 Mpb) from the start of their target gene. Mutations within them can disrupt sequence motifs for sequence-specific transcription factors, chromatin regulators and nucleosome positioning signals.

Examples of databases with non-coding/regulatory element information

- ENCODE (Encyclopedia of DNA Elements)
- Roadmap Epigenomics Project
- GTEx
- Others...

ENCODE

<https://www.encodeproject.org>



← Samples →

[illegible]

Roadmap

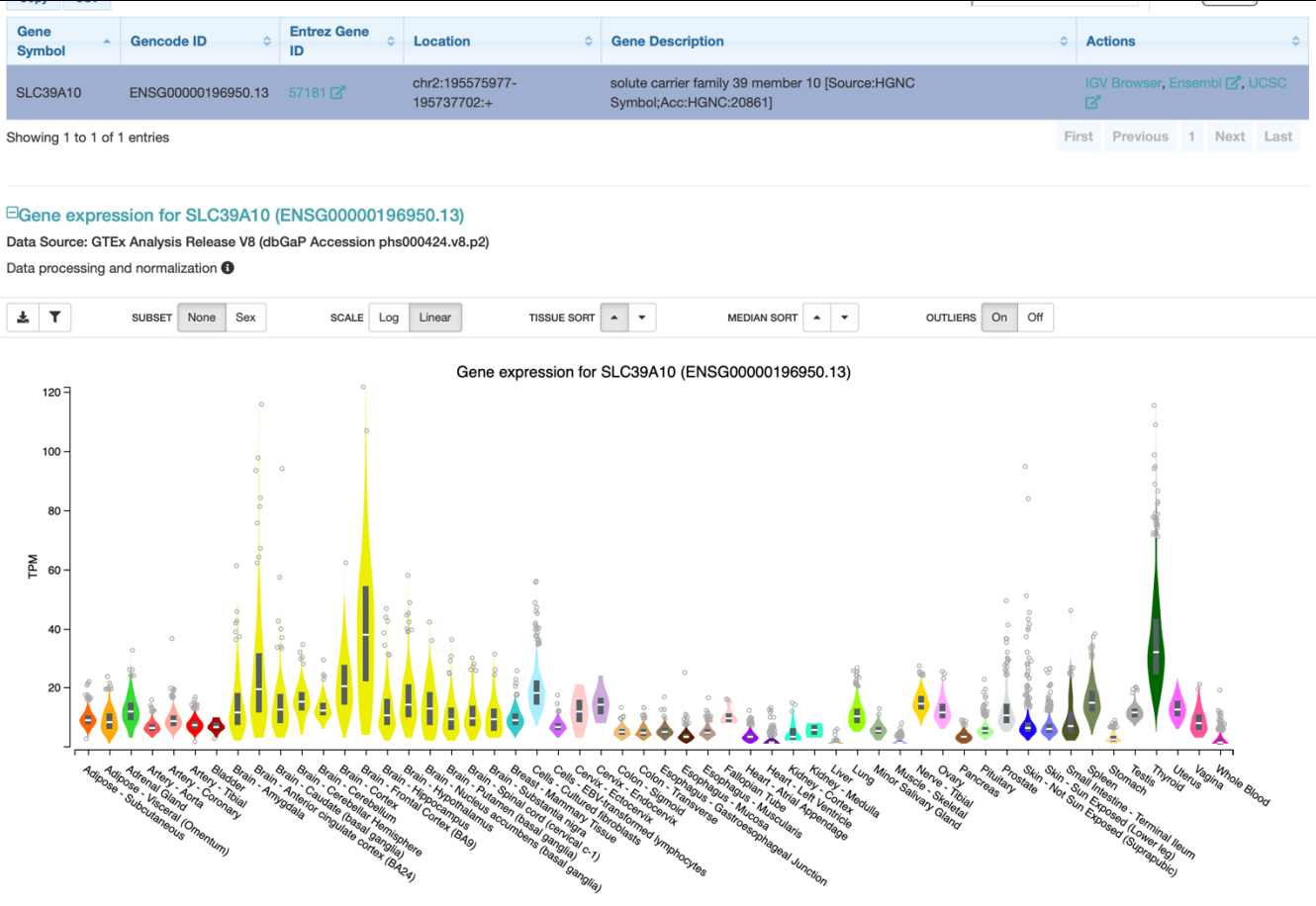


<https://gtexportal.org/>

[illegible]

GTEX

I searched “SLC39A10”



Examples of tools

- HaploReg
- RegulomeDB
- GTEx (portal)
- Genome browser(s)
- FUMA

HaploReg

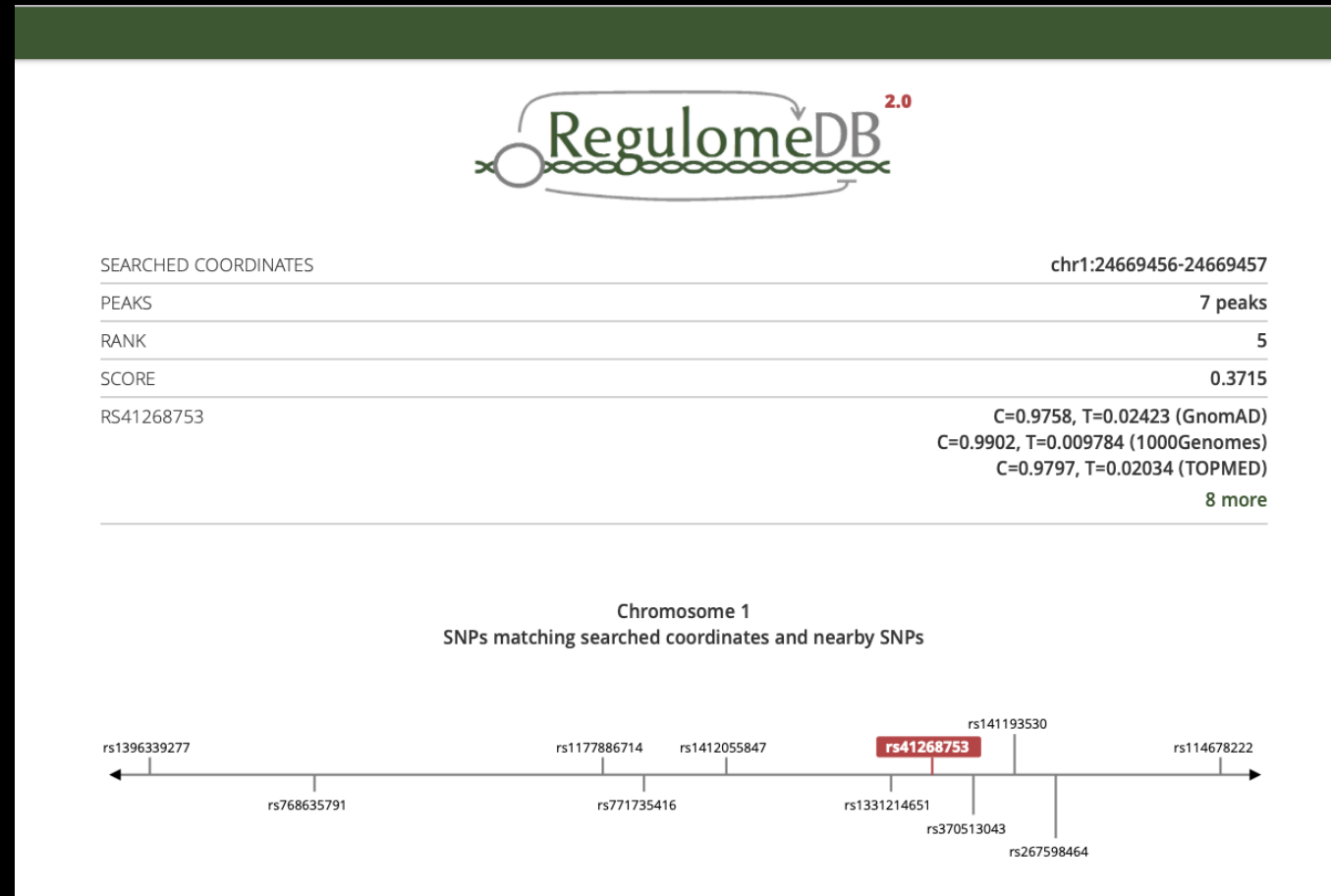
<https://pubs.broadinstitute.org/mammals/haploreg/haploreg.php>

I searched “rs12345”

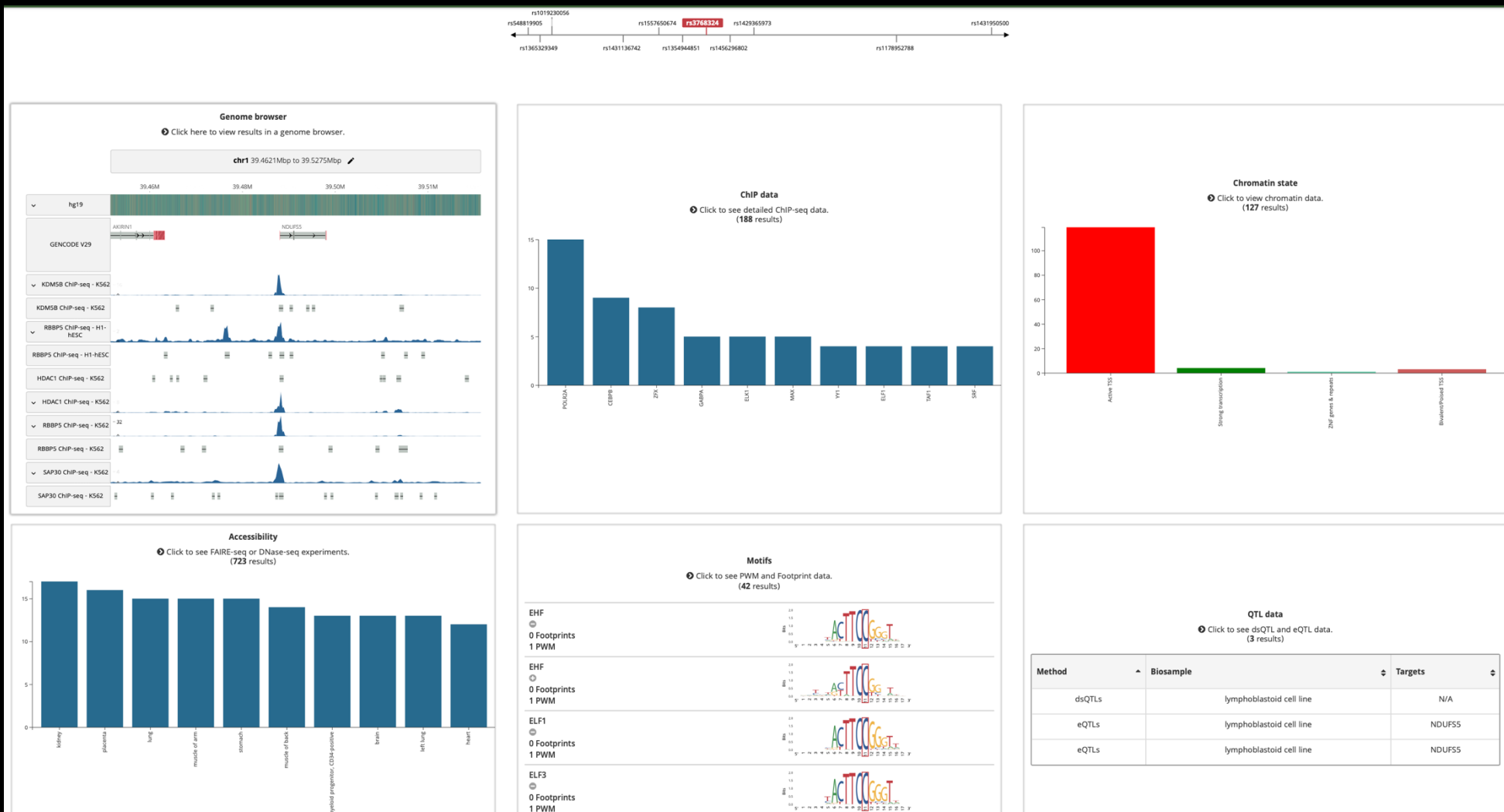
HaploReg v4.1																					
HaploReg is a tool for exploring annotations of the noncoding genome at variants on haplotype blocks, such as candidate regulatory SNPs at disease-associated loci. Using LD information from the 1000 Genomes Project, linked SNPs and small indels can be visualized along with chromatin state and protein binding annotation from the Roadmap Epigenomics and ENCODE projects, sequence conservation across mammals, the effect of SNPs on regulatory motifs, and the effect of SNPs on expression from eQTL studies. HaploReg is designed for researchers developing mechanistic hypotheses of the impact of non-coding variants on clinical phenotypes and normal variation. Update 2015.11.05: Version 4.1 GWAS and eQTL have been updated; a simpler pruning strategy is applied when combining GWAS; and links out to other NHGR/EBI GWAS hits and GRASP QTL hits are provided. Update 2015.09.15: Version 4.0 now includes many recent eQTL results including the GTEx pilot, four different options for defining enhancers using Roadmap Epigenomics data, and a complete set of source files for download and local analysis. Older versions available: v3, v2, v1. Build Query Set Options Documentation Use one of the three methods below to enter a set of variants. If an r^2 threshold is specified (see the Set Options tab), results for each variant will be shown in a separate table along with other variants in LD. If r^2 is set to NA, only queried variants will be shown, together in one table. Query (comma-delimited list of rsIDs OR a single region as chr:chr:start-end): rs12345 or, upload a text file (one refSNP ID per line): upload file selected or, select a GWAS:																					
Query SNP: rs12345 and variants with $r^2 \geq 0.8$																					
chr	pos (hg38)	r ²	D'	variant	Ref	Alt	AFR	AMR	ASN	EUR	SiPhy	Promoter	Enhancer	DNase	Proteins	Motifs	NHGR/EBI	GRASP	Selected eQTL	GENCODE	dbSNP
							freq	freq	freq	freq	cons	histone marks	histone marks		bound	hits	GWAS hits	hits	changed	genes	func annot
22	25432389	0.81	0.92	rs35876887	T	C	0.18	0.05	0.00	0.03						11 altered motifs			16 hits	27kb 5' of LRPLSL	
22	25432725	0.81	0.92	rs11090390	G	A	0.16	0.05	0.00	0.03			BLD			11 altered motifs			17 hits	27kb 5' of LRPLSL	
22	25433117	0.81	0.92	rs11090391	G	A	0.17	0.05	0.00	0.03			BLD			11 altered motifs			16 hits	28kb 5' of LRPLSL	
22	25433357	0.81	0.92	rs11090393	T	A/C	0.16	0.04	0.00	0.03			BLD						16 hits	28kb 5' of LRPLSL	
22	25436039	1	1	rs11704727	T	C	0.07	0.06	0.00	0.04						E2A,TAL1			20 hits	31kb 5' of LRPLSL	
22	25436168	0.92	1	rs201108613	AT	A	0.08	0.07	0.01	0.03						20 altered motifs			21 hits	31kb 5' of LRPLSL	
22	25436169	1	1	rs11704728	T	A	0.10	0.06	0.00	0.04						20 altered motifs			19 hits	31kb 5' of LRPLSL	
22	25438262	0.92	1	rs9620529	G	T	0.10	0.06	0.00	0.03			ESC, ESDR, ADRL			AIRE,EBF			19 hits	33kb 5' of LRPLSL	
22	25439230	1	1	rs11703485	G	A	0.14	0.07	0.00	0.04			BLD			5 altered motifs			18 hits	34kb 5' of LRPLSL	
22	25440661	1	1	rs76432327	A	G	0.18	0.08	0.00	0.04						Mat2,TATA			22 hits	35kb 5' of LRPLSL	
22	25444923	0.96	1	rs73408613	C	T	0.10	0.07	0.00	0.03						HNFI,Pos22			18 hits	40kb 5' of LRPLSL	
22	25448758	0.92	0.96	rs149613341	C	CG	0.09	0.06	0.00	0.04						NF-kappaB			19 hits	43kb 5' of LRPLSL	intronic
22	25451218	1	1	rs7287882	A	G	0.09	0.06	0.00	0.04						4 altered motifs			18 hits	46kb 5' of LRPLSL	intronic
22	25455184	1	1	rs9624838	C	G	0.09	0.06	0.00	0.04						4 altered motifs			20 hits	50kb 5' of LRPLSL	intronic
22	25455603	0.96	1	rs7284802	C	T	0.08	0.06	0.00	0.04						Nanog			20 hits	50kb 5' of LRPLSL	intronic
22	25456740	0.96	1	rs117035714	A	G	0.03	0.05	0.00	0.03			LIV		POL2			18 hits	51kb 5' of LRPLSL	intronic	
22	25456910	1	1	rs56384683	G	A	0.09	0.06	0.00	0.04			IPSC, LIV	IPSC	POL2			20 hits	52kb 5' of LRPLSL	intronic	
22	25456972	1	1	rs60807896	C	T	0.09	0.06	0.00	0.04			IPSC, LIV			GATA			20 hits	53kb 5' of LRPLSL	intronic
22	25457941	1	1	rs11704483	G	A	0.09	0.06	0.00	0.04								20 hits	53kb 5' of LRPLSL	intronic	
22	25458492	1	1	rs12345	G	A	0.09	0.06	0.00	0.04						Sp3Ax-20			19 hits	54kb 5' of LRPLSL	
22	25459983	1	1	rs11703557	A	G	0.09	0.06	0.00	0.04						Mat,NRSP,SETDB1			19 hits	55kb 5' of LRPLSL	intronic
22	25460289	1	1	rs11703617	A	G	0.09	0.06	0.00	0.04			BLD			8 altered motifs			20 hits	56kb 5' of LRPLSL	intronic
22	25461376	1	1	rs72084619	CAG	C	0.09	0.06	0.01	0.04			BLD			6 altered motifs			19 hits	56kb 5' of LRPLSL	
22	25462187	1	1	rs112613397	A	G	0.09	0.06	0.00	0.04						AP-2,Pax-5			19 hits	57kb 5' of LRPLSL	
22	25463804	0.96	1	rs9620531	A	T	0.02	0.04	0.00	0.03								18 hits	58kb 5' of LRPLSL		
22	25464263	0.9	1	rs146696411	G	GT	0.13	0.07	0.00	0.04						25 altered motifs			19 hits	59kb 5' of LRPLSL	
22	25472274	0.92	0.96	rs11703229	C	T	0.09	0.06	0.00	0.04						PRDM1,Rad21,VDR			18 hits	60kb 3' of AL008721.1	
22	25472602	0.92	0.96	rs11090395	G	A	0.08	0.06	0.00	0.04			SKIN			8 altered motifs			19 hits	60kb 3' of AL008721.1	
22	25473570	0.92	0.96	rs96205827	T	C	0.14	0.07	0.00	0.04						5 altered motifs			18 hits	60kb 3' of AL008721.1	
22	25473669	0.92	0.96	rs58892524	G	A	0.09	0.06	0.00	0.04						11 altered motifs			19 hits	59kb 3' of AL008721.1	
22	25474502	0.89	0.96	rs11090396	G	A	0.03	0.04	0.00	0.03			ADRL,LNG,GI			ERalpha-a			17 hits	58kb 3' of AL008721.1	
22	25476723	0.96	1	rs73410917	A	G	0.13	0.07	0.00	0.03			GI			4 altered motifs			18 hits	56kb 3' of AL008721.1	
22	25482096	0.92	0.96	rs10463120	A	G	0.14	0.07	0.00	0.04			ADRL					19 hits	56kb 3' of AL008721.1		
22	25491121	0.92	0.96	rs12168476	A	G	0.13	0.07	0.00	0.04						5 altered motifs			18 hits	42kb 3' of AL008721.1	
22	25491816	0.82	0.92	rs11705009	C	G	0.13	0.07	0.00	0.04						6 altered motifs			18 hits	41kb 3' of AL008721.1	
22	25493190	0.92	0.96	rs11109038	T	C	0.13	0.07	0.00	0.04						GR			18 hits	36kb 3' of AL008721.1	
22	25497117	0.92	0.96	rs9624842	C	T	0.13	0.07	0.00	0.04						CEBPB			19 hits	36kb 3' of AL008721.1	
22	25499395	0.83	0.96	rs9624843	C	T	0.34	0.09	0.10	0.04						ERalpha-a,lrf			26 hits	33kb 3' of AL008721.1	
22	25501736	0.89	0.96	rs12155353	G	T	0.08	0.06	0.00	0.03			BRST,STRM			4 altered motifs			19 hits	31kb 3' of AL008721.1	
22	25502047	0.89	0.96	rs199778348	G	GT	0.07	0.06	0.00	0.03			BRST,BLD,SKIN			4 altered motifs			18 hits	30kb 3' of AL008721.1	
22	25502252	0.89	0.96	rs19995930	AT	A	0.07	0.06	0.00	0.03			BRST,BLD,SKIN			4 altered motifs			19 hits	30kb 3' of AL008721.1	
22	25503972	0.85	0.92	rs6004884	A	C	0.38	0.08	0.00	0.04						7 altered motifs		1 hit	27 hits	29kb 3' of AL008721.1	
22	25504219	0.89	0.96	rs9624847	G	A	0.11	0.07	0.00	0.03			ADRL		4 issues	Esrr2,PRXA			22 hits	29kb 3' of AL008721.1	
22	25504819	0.81	0.92	rs111325370	ESC	C	0.08	0.06	0.00	0.03			ESC			4 altered motifs			20 hits	28kb 3' of AL008721.1	
22	25509253	0.81	0.92	rs12169096	A	G	0.08	0.06	0.00	0.03			5 issues			BDP1			19 hits	23kb 3' of AL008721.1	
22	25510312	0.81	0.92	rs742031	C	T	0.08	0.06	0.00	0.03			ESC,ESDR,BRST		OVRY	CEBPB,Pax-5,p300			19 hits	22kb 3' of AL008721.1	
22	25514348	0.85	0.96	rs7286386	C	T	0.10	0.06	0.00	0.03						Myf			26 hits	16kb 3' of AL008721.1	
22	25516849	0.81	0.92	rs9620543	C	T	0.10	0.06	0.00	0.03						HSF2			17 hits	16kb 3' of AL008721.1	
22	25518203	0.81	0.92	rs12168294	T	C	0.23	0.07	0.00	0.03			LIV	LIV	4 bound proteins	BRCA1,COMP1,Spdel			21 hits	15kb 3' of AL008721.1	
22	25519660	0.81	0.92	rs7284167	T	C	0.23	0.07	0.00	0.03			ESDR,LIV			STAT,TCF4			24 hits	13kb 3' of AL008721.1	
22	25520419	0.81	0.92	rs7280650	A	G	0.23	0.07	0.00	0.03						ESR,STAT,TAL1			24 hits	13kb 3' of AL008721.1	
22	25521181	0.81	0.92	rs7284654	G	T	0.24	0.07	0.00	0.03								23 hits	12kb 3' of AL008721.1		

RegulomeDB

<https://regulomedb.org/regulome-search/>



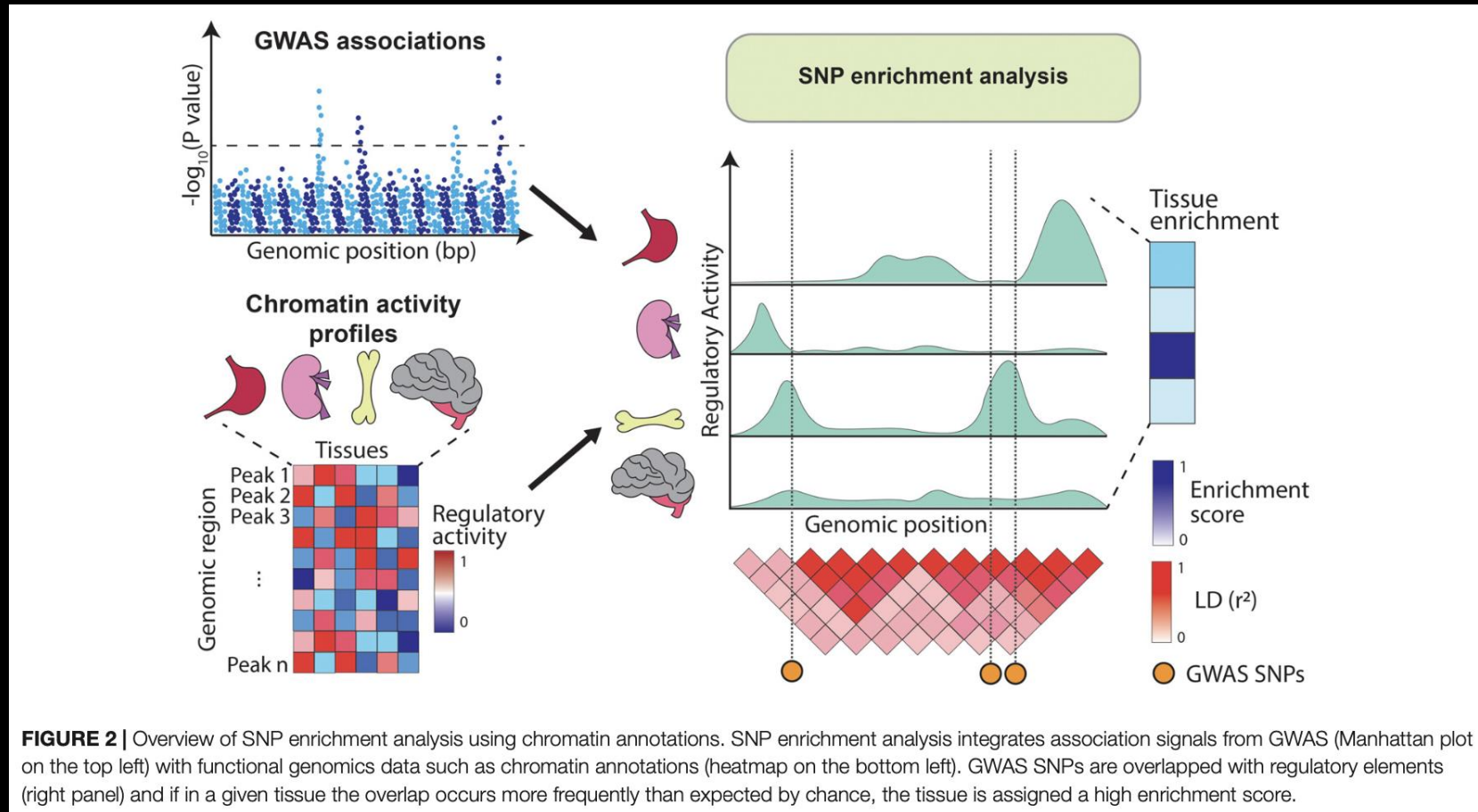
RegulomeDB



“Association-based” annotation methods

- Examples of information of interest
 - Are the loci associated with my trait enriched for...
 - Expression in a tissue/cell type? At a particular developmental stage?
 - Particular chromatin markings?
 - Known regulatory elements for a particular tissue/cell type/stage?
 - Genes in a network? Pathway? Other categories of genes?
- Examples of some enrichment databases/tools
 - HaploReg
 - FUMA
 - A lot of types of gene annotation/gene sets exist

Enrichment – main idea



Example: enrichment testing in HaploReg

HaploReg v4.1



HaploReg is a tool for exploring annotations of the noncoding genome at variants on haplotype blocks, such as candidate regulatory SNPs at disease-associated loci. Using LD information from the 1000 Genomes Project, linked SNPs and small indels can be visualized along with chromatin state and protein binding annotation from the Roadmap Epigenomics and ENCODE projects, sequence conservation across mammals, the effect of SNPs on regulatory motifs, and the effect of SNPs on expression from eQTL studies. HaploReg is designed for researchers developing mechanistic hypotheses of the impact of non-coding variants on clinical phenotypes and normal variation.

Update 2015.11.05: Version 4.1 GWAS and eQTL have been updated; a simpler pruning strategy is applied when combining GWAS; and links out to other NHGRI/EBI GWAS hits and GRASP QTL hits are provided.

Update 2015.09.15: Version 4.0 now includes many recent eQTL results including the GTEx pilot, four different options for defining enhancers using Roadmap Epigenomics data, and a complete set of source files for download and local analysis. Older versions available: [v3](#), [v2](#), [v1](#).

[Build Query](#) [Set Options](#) [Documentation](#)

Use one of the three methods below to enter a set of variants. If an r^2 threshold is specified (see the Set Options tab), results for each variant will be shown in a separate table along with other variants in LD. If r^2 is set to NA, only queried variants will be shown, together in one table.

Query (comma-delimited list of rsIDs OR a single region as chrN:start-end):

or, upload a text file (one refSNP ID per line): Choose File no file selected

or, select a GWAS:

Submit

Are these 18 lupus associations enriched for enhancers in particular cell types?

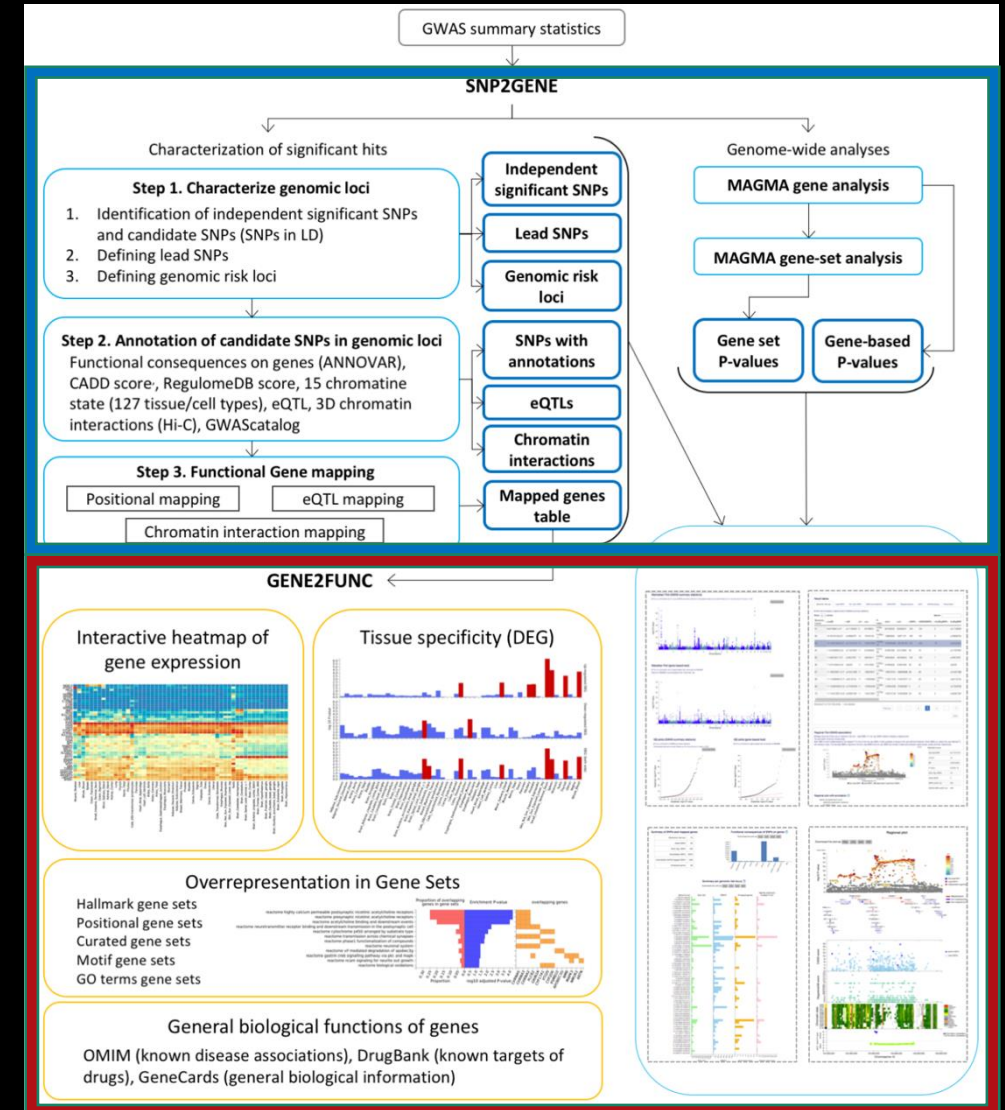
Example: Enrichment testing in HaploReg

Cell	Observed	Expected (all SNPs)	Expected (GWAS SNPs)	Binomial p (all SNPs)	Binomial p (GWAS SNPs)
E017 LNG.IMR90 (IMR90 fetal lung fibroblasts Cell Line)	1	0.8	1.5	0.553887	0.784606
E002 ESC.WA7 (ES-WA7 Cells)	1	0.2	0.4	0.220089	0.353194
E008 ESC.H9 (H9 Cells)	1	0.3	0.6	0.264516	0.432262
E001 ESC.I3 (ES-I3 Cells)	0	0.7	1.1	1	1
E015 ESC.HUES6 (HUES6 Cells)	0	0.7	1.1	1	1
E014 ESC.HUES48 (HUES48 Cells)	0	0.7	1	1	1
E016 ESC.HUES64 (HUES64 Cells)	0	0.6	1	1	1
E003 ESC.H1 (H1 Cells)	1	0.6	1	0.445536	0.648016
E024 ESC.4STAR (ES-UCSF4 Cells)	0	0.7	1.2	1	1
E020 IPSC.20B (IPS-20b Cells)	0	0.5	0.8	1	1
E019 IPSC.18 (IPS-18 Cells)	1	0.7	1	0.492131	0.634069
E018 IPSC.15b (IPS-15b Cells)	1	0.6	1	0.473802	0.644263
E021 IPSC.DF.6.9 (IPS DF 6.9 Cells)	2	0.3	0.6	0.034322	0.128701
E022 IPSC.DF.19.11 (IPS DF 19.11 Cells)	1	0.7	1.3	0.503375	0.734391
E007 ESDR.H1.NEUR.PROG (H1 Derived Neuronal Progenitor Cultured Cells)	0	0.4	0.7	1	1
E009 ESDR.H9.NEUR.PROG (H9 Derived Neuronal Progenitor Cultured Cells)	0	0.5	0.9	1	1
E010 ESDR.H9.NEUR (H9 Derived Neuron Cultured Cells)	0	0.6	1.1	1	1
E013 ESDR.CD56.MESO (hESC Derived CD56+ Mesoderm Cultured Cells)	0	0.6	1.1	1	1
E012 ESDR.CD56.ECTO (hESC Derived CD56+ Ectoderm Cultured Cells)	1	0.5	1	0.41696	0.6236
E011 ESDR.CD184.ENDO (hESC Derived CD184+ Endoderm Cultured Cells)	0	0.6	1.1	1	1
E004 ESDR.H1.BMP4.MESO (H1 BMP4 Derived Mesendoderm Cultured Cells)	0	0.3	0.5	1	1
E005 ESDR.H1.BMP4.TROP (H1 BMP4 Derived Trophoblast Cultured Cells)	1	0.6	1.3	0.482068	0.737239
E006 ESDR.H1.MSC (H1 Derived Mesenchymal Stem Cells)	1	0.7	1.3	0.510752	0.734391
E062 BLD.PER.MONUC.PC (Primary mononuclear cells from peripheral blood)	3	0.2	0.5	0.001142	0.011283
E034 BLD.CD3.PPC (Primary T cells from peripheral blood)	4	0.6	1	0.003117	0.016507
E045 BLD.CD4.CD251.CD127.TMEMPC (Primary T cells effector/memory enriched from peripheral blood)	3	0.3	0.5	0.002326	0.012604
E033 BLD.CD3.CPC (Primary T cells from cord blood)	1	0.4	0.7	0.361645	0.520797
E044 BLD.CD4.CD25.CD127M.TREGPC (Primary T regulatory cells from peripheral blood)	1	0.4	0.7	0.301398	0.484582
E043 BLD.CD4.CD25M.TPC (Primary T helper cells from peripheral blood)	4	0.6	1	0.002706	0.015267
E039 BLD.CD4.CD25M.CD45RA.NPC (Primary T helper naive cells from peripheral blood)	2	0.5	0.9	0.101566	0.20945
E041 BLD.CD4.CD25M.IL17M.PL.TPC (Primary T helper cells PMA-I stimulated)	6	0.6	1	2.5E-5	0.000285
E042 BLD.CD4.CD25M.IL17P.PL.TPC (Primary T helper 17 cells PMA-I stimulated)	1	0.5	0.9	0.385566	0.597591
E040 BLD.CD4.CD25M.CD45RO.MPC (Primary T helper memory cells from peripheral blood 1)	4	0.5	0.9	0.0015	0.010103
E037 BLD.CD4.MPC (Primary T helper memory cells from peripheral blood 2)	3	0.6	1	0.018899	0.06928
E048 BLD.CD8.MPC (Primary T CD8+ memory cells from peripheral blood)	3	0.4	0.8	0.009554	0.042671
E038 BLD.CD4.NPC (Primary T helper naive cells from peripheral blood)	3	0.5	0.8	0.011441	0.041778
E047 BLD.CD8.NPC (Primary T CD8+ naive cells from peripheral blood)	3	0.5	1	0.015628	0.06928
E029 BLD.CD14.PC (Primary monocytes from peripheral blood)	5	0.8	1.4	0.000821	0.00977
E031 BLD.CD19.CPC (Primary B cells from cord blood)	5	0.5	0.8	8.9E-5	0.001122
E035 BLD.CD34.PC (Primary hematopoietic stem cells)	3	0.5	1	0.01456	0.06928
E051 BLD.MOB.CD34.PC.M (Primary hematopoietic stem cells G-CSF-mobilized Male)	1	0.7	1.2	0.51805	0.71971
E050 BLD.MOB.CD34.PC.F (Primary hematopoietic stem cells G-CSF-mobilized Female)	3	0.7	1.3	0.036417	0.142348
E036 BLD.CD34.CC (Primary hematopoietic stem cells short term culture)	3	0.7	1.3	0.029167	0.126309
E032 BLD.CD19.PPC (Primary B cells from peripheral blood)	5	0.7	1.2	0.000501	0.005631
E046 BLD.CD56.PC (Primary Natural Killer cells from peripheral blood)	4	0.6	1.1	0.002908	0.019783
E030 BLD.CD15.PC (Primary neutrophils from peripheral blood)	2	0.5	0.9	0.09755	0.220618
E026 STRM.MRW.MSC (Bone Marrow Derived Cultured Mesenchymal Stem Cells)	2	0.6	1.1	0.130625	0.312602
E049 STRM.CHON.MRW.DR.MSC (Mesenchymal Stem Cell Derived Chondrocyte Cultured Cells)	2	0.8	1.4	0.171399	0.416692
E025 FAT.ADIP.DR.MSC (Adipose Derived Mesenchymal Stem Cell Cultured Cells)	3	1	1.7	0.0795	0.241757
E023 FAT.MSC.DR.ADIP (Mesenchymal Stem Cell Derived Adipocyte Cultured Cells)	1	0.8	1.3	0.543425	0.738182
E052 MUS.SAT (Muscle Satellite Cultured Cells)	3	0.7	1.2	0.032384	0.12482
E055 SKIN.PEN.FRSK.FIB.01 (Foreskin Fibroblast Primary Cells skin01)	2	0.8	1.4	0.179289	0.415486
E056 SKIN.PEN.FRSK.FIB.02 (Foreskin Fibroblast Primary Cells skin02)	2	0.5	1	0.098195	0.274672
E059 SKIN.PEN.FRSK.MEL.01 (Foreskin Melanocyte Primary Cells skin01)	3	0.4	0.8	0.008948	0.048716
E061 SKIN.PEN.FRSK.MEL.03 (Foreskin Melanocyte Primary Cells skin03)	3	0.8	1.2	0.039991	0.117484
E057 SKIN.PEN.FRSK.KER.02 (Foreskin Keratinocyte Primary Cells skin02)	1	0.7	1.2	0.509326	0.723698
E058 SKIN.PEN.FRSK.KER.03 (Foreskin Keratinocyte Primary Cells skin03)	2	0.7	1.3	0.166327	0.360279
E028 BRST.HMEC.35 (Breast variant Human Mammary Epithelial Cells (vHMEC))	2	0.8	1.4	0.179461	0.404587

FUMA (Functional Mapping and Annotation of GWAS)

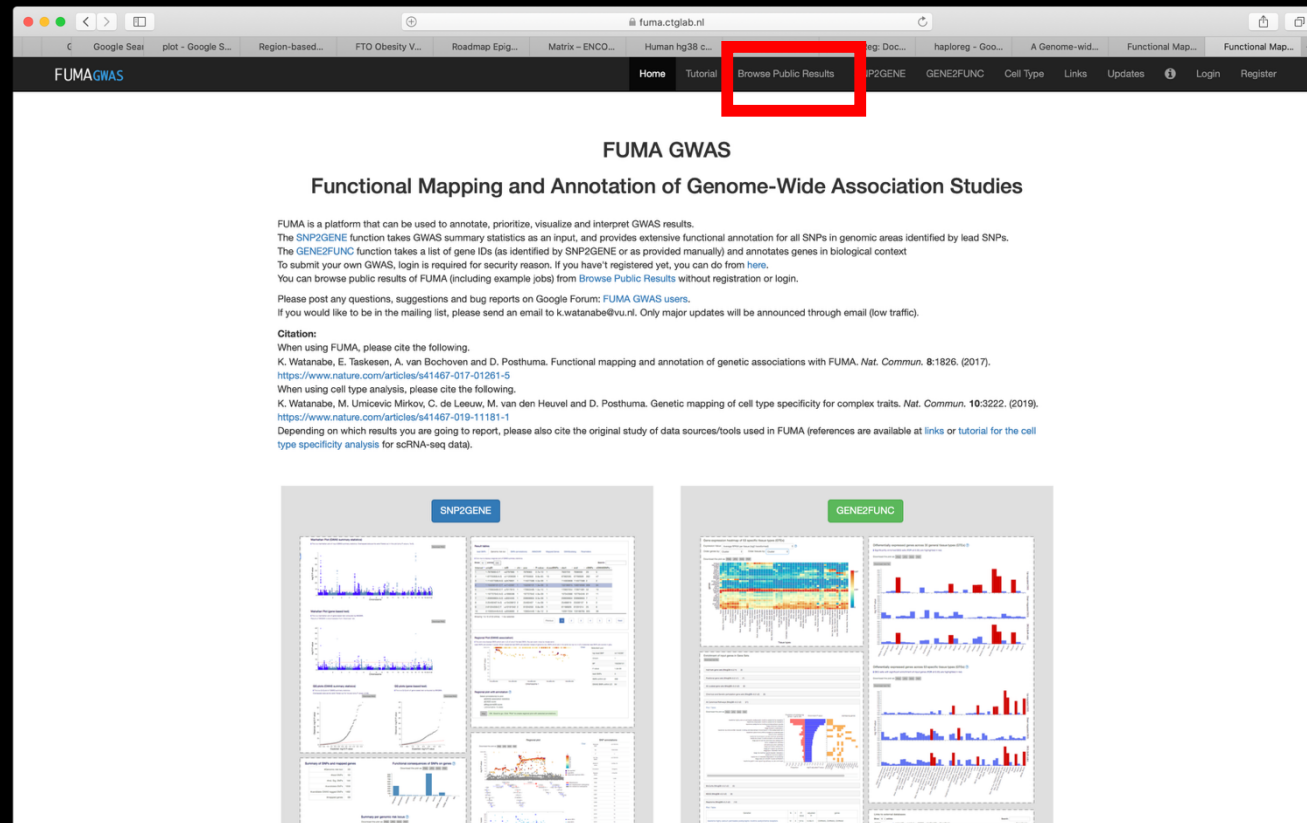
<https://fuma.ctglab.nl>

1. Prioritize genes from the various loci



2. Gene enrichment tests, etc.

FUMA annotation example



The screenshot shows the FUMA GWAS website interface. The browser address bar displays `fuma.ctglab.nl`. The navigation bar includes links for Home, Tutorial, **Browse Public Results** (highlighted with a red box), SNP2GENE, GENE2FUNC, Cell Type, Links, Updates, Login, and Register. The main heading is "FUMA GWAS" followed by the subtitle "Functional Mapping and Annotation of Genome-Wide Association Studies".

FUMA is a platform that can be used to annotate, prioritize, visualize and interpret GWAS results. The **SNP2GENE** function takes GWAS summary statistics as an input, and provides extensive functional annotation for all SNPs in genomic areas identified by lead SNPs. The **GENE2FUNC** function takes a list of gene IDs (as identified by SNP2GENE or as provided manually) and annotates genes in biological context. To submit your own GWAS, login is required for security reason. If you haven't registered yet, you can do from [here](#). You can browse public results of FUMA (including example jobs) from [Browse Public Results](#) without registration or login.

Please post any questions, suggestions and bug reports on Google Forum: [FUMA GWAS users](#). If you would like to be in the mailing list, please send an email to k.watanabe@vu.nl. Only major updates will be announced through email (low traffic).

Citation:
When using FUMA, please cite the following.
K. Watanabe, E. Taskesen, A. van Bochoven and D. Posthuma. Functional mapping and annotation of genetic associations with FUMA. *Nat. Commun.* 8:1826. (2017). <https://www.nature.com/articles/s41467-017-01261-5>
When using cell type analysis, please cite the following.
K. Watanabe, M. Umicevic Mirkov, C. de Leeuw, M. van den Heuvel and D. Posthuma. Genetic mapping of cell type specificity for complex traits. *Nat. Commun.* 10:3222. (2019). <https://www.nature.com/articles/s41467-019-11181-1>
Depending on which results you are going to report, please also cite the original study of data sources/tools used in FUMA (references are available at [links](#) or [tutorial](#) for the cell type specificity analysis for scRNA-seq data).

Below the text, there are two preview panels. The left panel, labeled "SNP2GENE", shows a genomic track with various annotations. The right panel, labeled "GENE2FUNC", shows a heatmap of gene expression across different cell types, with a bar chart on the right side.

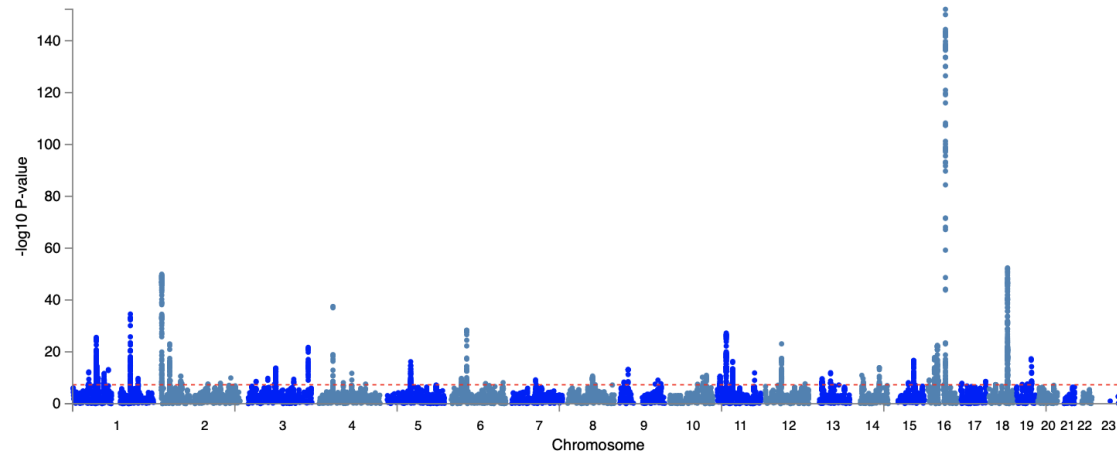
FUMA annotation example

Manhattan Plot (GWAS summary statistics)

i Manhattan plot of the input GWAS summary statistics.

For plotting, overlapping data points are not drawn (filtering was performed only for SNPs with P-value $\geq 1e-5$, see tutorial for details).

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



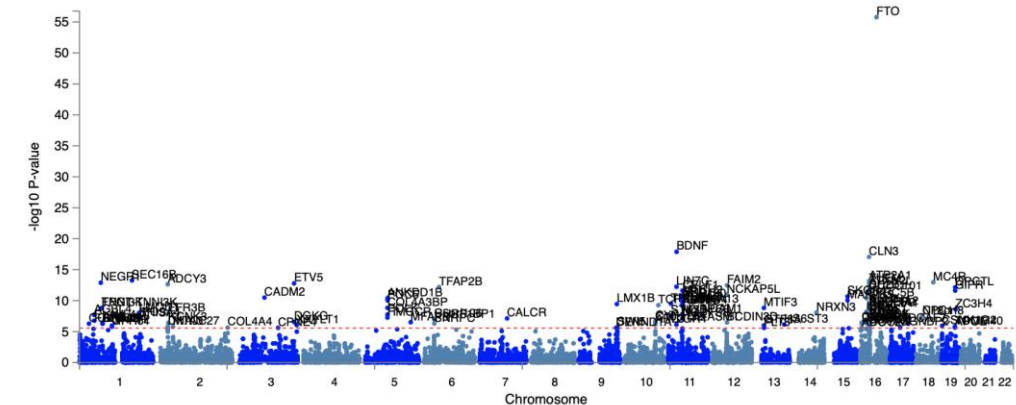
Mahattan Plot (gene-based test)

i This is a manhattan plot of the gene-based test as computed by MAGMA based on your input GWAS summary statistics. The gene-based P-value is downloadable from 'Download' tab from the left side bar.

Input SNPs were mapped to 17794 protein coding genes. Genome wide significance (red dashed line in the plot) was defined at $P = 0.05/17794 = 2.810e-6$.

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)

Label top genes.

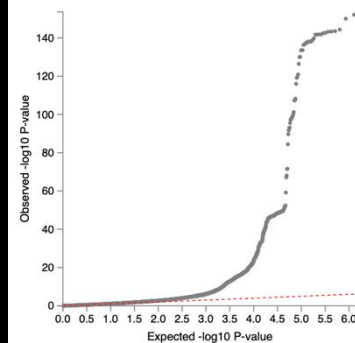


QQ plot (GWAS summary statistics)

i This is a Q-Q plot of GWAS summary statistics.

For plotting purposes, overlapping data points are not drawn (filtering was performed only for SNPs with P-value $\geq 1e-5$, see tutorial for details).

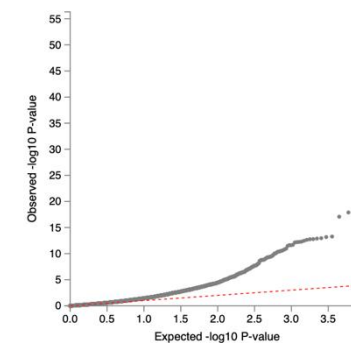
Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



QQ plot (gene-based test)

i This is a Q-Q plot of the gene-based test computed by MAGMA.

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



Fuma annotation example

Find the “independent” loci and summarize them

Result tables

Genomic risk loci | **lead SNPs** | Ind. Sig. SNPs | SNPs (annotations) | ANNOVAR | Mapped Genes | GWAScatalog

Parameters

i Click row to display a regional plot of GWAS summary statistics.

Show **10** entries Search:

Genomic Locus	uniqID	rsID	chr	pos	P-value	start	end	nSNPs	nGWASSNPs	nIndSigSNPs
1	1:7879063:C:T	rs2797685	1	7879063	2.7e-10	7822723	7907971	29	5	1
2	1:67705958:A:G	rs11209026	1	67705958	9.9e-65	67362595	67766926	291	47	13
3	1:114377568:A:G	rs2476601	1	114377568	4.5e-09	114303808	114377568	2	2	1
4	1:155230131:C:T	rs1142287	1	155230131	1.3e-09	155198347	156016356	380	64	3
5	1:172853460:C:T	rs7517810	1	172853460	1.5e-15	172803958	172871681	59	10	1
6	1:197727642:A:G	rs1998598	1	197727642	4.9e-09	197342686	197784249	66	11	1
7	1:206939904:A:G	rs3024505	1	206939904	8.3e-09	206939904	206968955	8	1	1
8	2:25492467:A:G	rs13428812	2	25492467	1.4e-08	25488819	25506107	9	2	1
9	2:61224259:C:T	rs10181042	2	61224259	6.6e-09	61186829	61231014	26	6	1
10	2:103054449:A:G	rs2058660	2	103054449	1.6e-12	102917234	103201364	341	39	3

Showing 1 to 10 of 52 entries

Previous **1** 2 3 4 5 6 Next

Find the lead SNPs and summarize them

Result tables

Genomic risk loci | **lead SNPs** | Ind. Sig. SNPs | SNPs (annotations) | ANNOVAR | Mapped Genes | GWAScatalog

Parameters

i Click row to display a regional plot of GWAS summary statistics.

Show **10** entries Search:

No	Genomic Locus	uniqID	rsID	chr	pos	P-value	nIndSigSNPs	IndSigSNPs
1	1	1:7879063:C:T	rs2797685	1	7879063	2.7e-10	1	rs2797685
2	2	1:67526029:A:G	rs12061545	1	67526029	1.8e-08	2	rs12061545;rs6656890
3	2	1:67627260:C:T	rs12041056	1	67627260	2.7e-08	6	rs12041056;rs11209018;rs11805303;rs120303
4	2	1:67681669:G:T	rs7517847	1	67681669	1.6e-53	7	rs7517847;rs1343151;rs12045232;rs1120903
5	2	1:67705958:A:G	rs11209026	1	67705958	9.9e-65	4	rs11209026;rs1343151;rs6656890;rs1203094
6	2	1:67744601:A:G	rs6660226	1	67744601	1.5e-22	5	rs6660226;rs11805303;rs1343151;rs1204523
7	3	1:114377568:A:G	rs2476601	1	114377568	4.5e-09	1	rs2476601
8	4	1:155230131:C:T	rs1142287	1	155230131	1.3e-09	3	rs1142287;rs821551;rs867549
9	5	1:172853460:C:T	rs7517810	1	172853460	1.5e-15	1	rs7517810
10	6	1:197727642:A:G	rs1998598	1	197727642	4.9e-09	1	rs1998598

Showing 1 to 10 of 75 entries

Previous **1** 2 3 4 5 ... 8 Next

Fuma annotation example

Find SNPs in LD with your associations and merge with annotation

Result tables

Genomic risk loci lead SNPs Ind. Sig. SNPs **SNPs (annotations)** ANNOVAR Mapped Genes GWAScatalog

Parameters

i This table contain all SNPs in LD of identified lead SNPs (functional filtering for gene mapping is not applied in this table).
i In this table "alt" is the risk allele if provided in the input GWAS file. See tutorial for details.

Show 10 entries Search:

uniqID	rsID	chr	pos	non_effect_allele	effect_allele	MAF	gwasP	Genomic Locus	r2	IndSig
10:101271789:A:C	rs11190126	10	101271789	A	C	0.3877	NA	32	0.987867	rs11190126
10:101271982:A:C	rs11190127	10	101271982	C	A	0.3678	NA	32	0.792402	rs11190127
10:101271995:C:CA	rs144809513	10	101271995	CA	C	0.3956	NA	32	0.931223	rs11190126
10:101273546:A:G	rs61872688	10	101273546	G	A	0.3837	NA	32	0.987626	rs11190126
10:101273857:C:T	rs10883358	10	101273857	C	T	0.3847	NA	32	0.991778	rs11190126
10:101274033:A:G	rs10883359	10	101274033	A	G	0.3022	NA	32	0.672971	rs11190126
10:101274058:C:T	rs7095491	10	101274058	C	T	0.4771	6.2e-20	32	0.922544	rs4409764
10:101274251:C:CT	rs11361434	10	101274251	CT	C	0.3847	NA	32	0.991778	rs11190126
10:101274365:A:G	rs7078219	10	101274365	A	G	0.3847	9.6e-16	32	0.991778	rs11190126
10:101274425:C:T	rs7095953	10	101274425	C	T	0.3022	2.9e-11	32	0.672971	rs11190126

Showing 1 to 10 of 8,717 entries

Previous 1 2 3 4 5 ... 872 Next

Result tables

Genomic risk loci lead SNPs Ind. Sig. SNPs **SNPs (annotations)** ANNOVAR Mapped Genes GWAScatalog

Parameters

i This table contain all SNPs in LD of identified lead SNPs (functional filtering for gene mapping is not applied in this table).
i In this table "alt" is the risk allele if provided in the input GWAS file. See tutorial for details.

Show 10 entries Search:

asP	Genomic Locus	r2	IndSigSNP	Nearest gene	dist	position	CADD	RDB	minChrState(127)	commonChrState(127)
✓	32	0.987867	rs11190134	RP11-129J12.2	14317	intergenic	0.145	7	13	14
✓	32	0.792402	rs11190134	RP11-129J12.2	14124	intergenic	4.585	7	13	14
✓	32	0.931223	rs11190134	RP11-129J12.2	14111	intergenic	1.771	NA	13	14
✓	32	0.987626	rs11190134	RP11-129J12.2	12560	intergenic	2.268	6	13	14
✓	32	0.991778	rs11190134	RP11-129J12.2	12249	intergenic	1.654	6	13	14
✓	32	0.672971	rs11190134	RP11-129J12.2	12073	intergenic	3.689	7	13	14
6e-20	32	0.922544	rs4409764	RP11-129J12.2	12048	intergenic	4.886	7	13	14
✓	32	0.991778	rs11190134	RP11-129J12.2	11855	intergenic	9.708	NA	9	14
6e-16	32	0.991778	rs11190134	RP11-129J12.2	11741	intergenic	3.778	5	9	14
6e-11	32	0.672971	rs11190134	RP11-129J12.2	11681	intergenic	7.084	5	9	14

Showing 1 to 10 of 8,717 entries

Previous 1 2 3 4 5 ... 872 Next

FUMA annotation example

Are your SNPS in GWAScatalog?

Result tables

[Genomic risk loci](#) [lead SNPs](#) [Ind. Sig. SNPs](#) [SNPs \(annotations\)](#) [ANNOVAR](#) [Mapped Genes](#) [GWAScatalog](#)

[Parameters](#)

i This table only shows subset of information from GWAS catalog.
Please download a output file (gwascatalog.txt) from "Download" tab to get full information

Show entries

Search:

Genomic Locus	IndSigSNP	chr	bp	rsID	PMID	Trait	FirstAuth	Date	P-value
1	rs2797685	1	7879063	rs2797685	21102463	Crohn's disease	Franke A	2010-11-21	7E-9
2	rs10889657	1	67600686	rs12044149	26626624	Psoriatic arthritis	Stuart PE	2015-11-28	3E-12
2	rs10889657	1	67611613	rs4655683	30552173	Psoriatic arthritis	Aterido A	2018-12-14	8E-7
2	rs7517847	1	67669634	rs11580078	26301688	Pediatric autoimmune diseases	Li YR	2015-08-24	8E-11
2	rs11805303	1	67675516	rs11805303	17554300	Crohn's disease	Wellcome Trust Case Control Consortium	2007-06-07	6E-12
2	rs7517847	1	67681669	rs7517847	17435756	Crohn's disease	Rioux JD	2007-04-15	3E-12
2	rs7517847	1	67681669	rs7517847	24823311	Plasma omega-6 polyunsaturated fatty acid levels (linoleic acid)	Guan W	2014-05-13	4E-6
2	rs7517847	1	67681669	rs7517847	26974007	Chronic inflammatory diseases (ankylosing spondylitis, Crohn's disease, psoriasis, primary sclerosing cholangitis, ulcerative colitis) (pleiotropy)	Ellinghaus D	2016-03-14	1E-98
2	rs7517847	1	67688349	rs10489629	30019117	Adolescent idiopathic scoliosis	Liu J	2018-07-17	5E-7
2	rs11805303	1	67694202	rs2201841	19169254	Psoriasis	Nair RP	2009-01-25	3E-8

Showing 1 to 10 of 1,547 entries

Previous

1

2

3

4

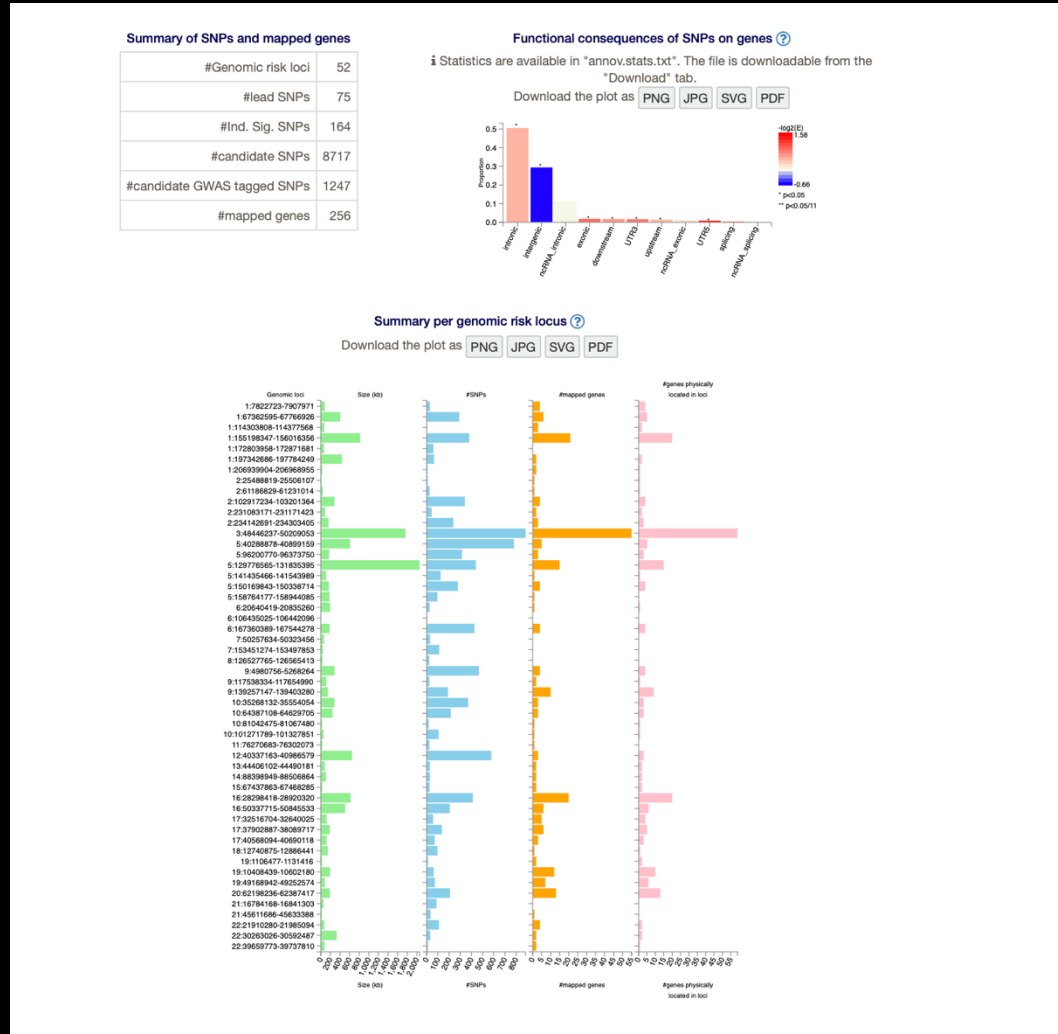
5

...

155

Next

FUMA annotation example



FUMA annotation example

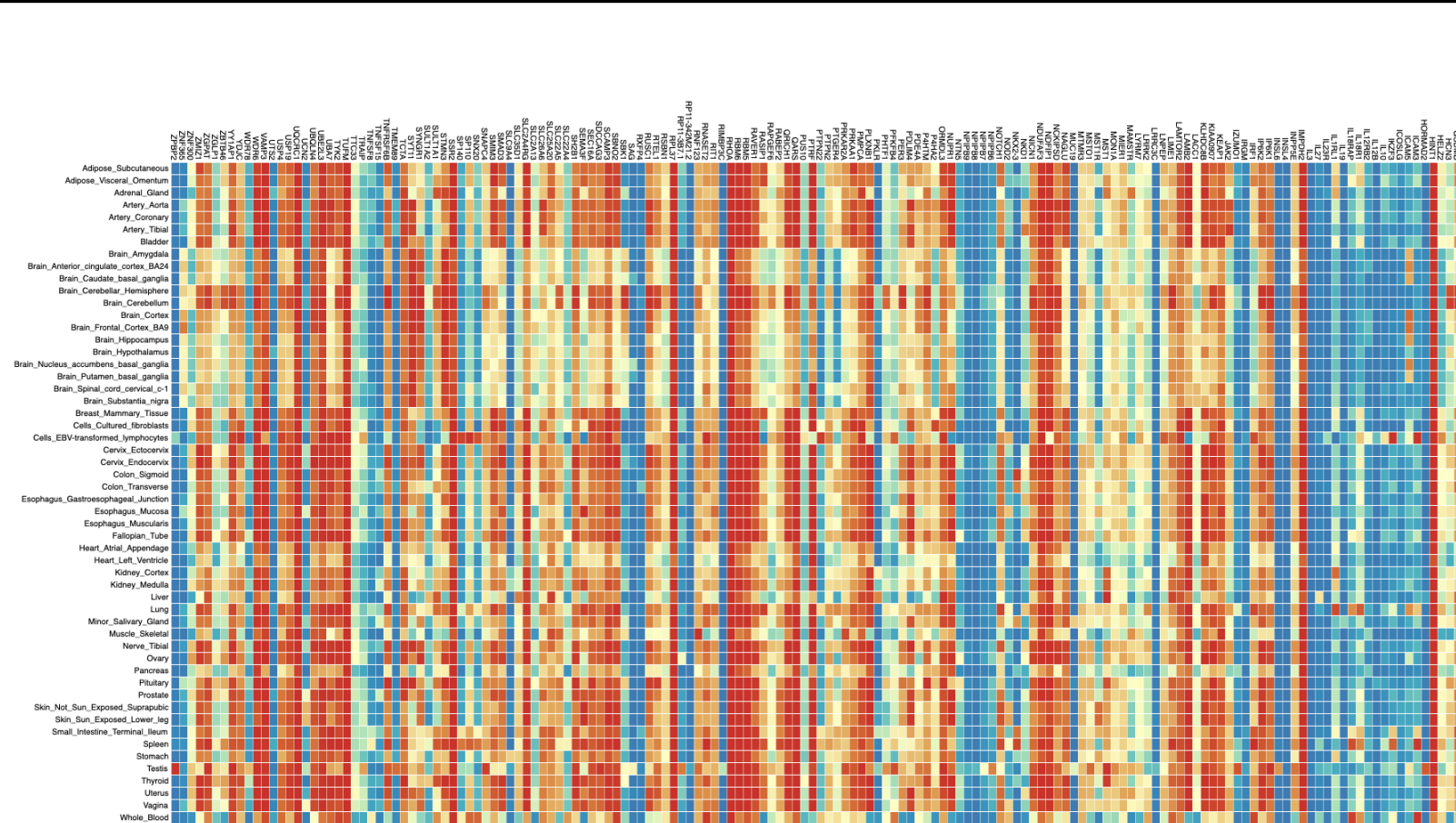
The first step mapped the GWAS associations to a set of genes.
The second step analyzes the gene set.

Summary of input genes

Number of input genes	256
Number of background genes	20260
Number of input genes with recognised Ensembl ID	256
Input genes without recognised Ensembl ID	NA
Number of background genes with recognised Ensembl ID	20119
Background genes without recognised Ensembl ID	NA
Number of input genes with unique entrez ID	242
Number of background genes with unique entrez ID	19142

FUMA annotation example

Expression of mapped genes.



Are the genes enriched for expression in certain tissue types?

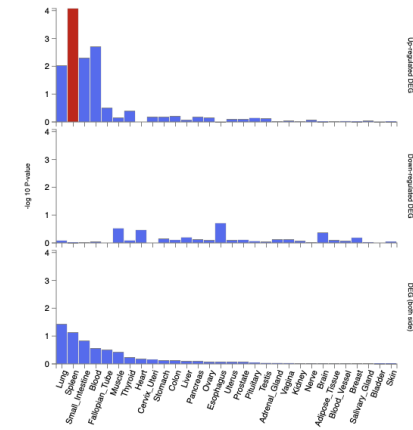
Differentially expressed genes

Significantly enriched DEG sets ($P_{\text{bon}} < 0.05$) are highlighted in red.

Order tissue by : two-side DEG P-value

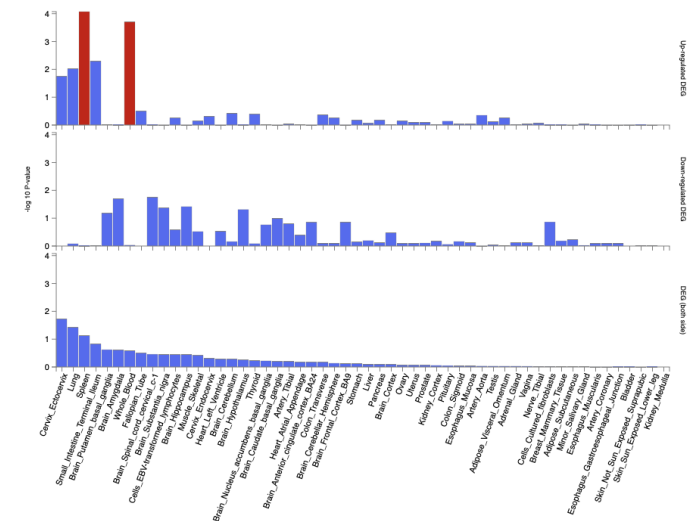
GTEx v8 30 general tissue types

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



GTEx v8 54 tissue types

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



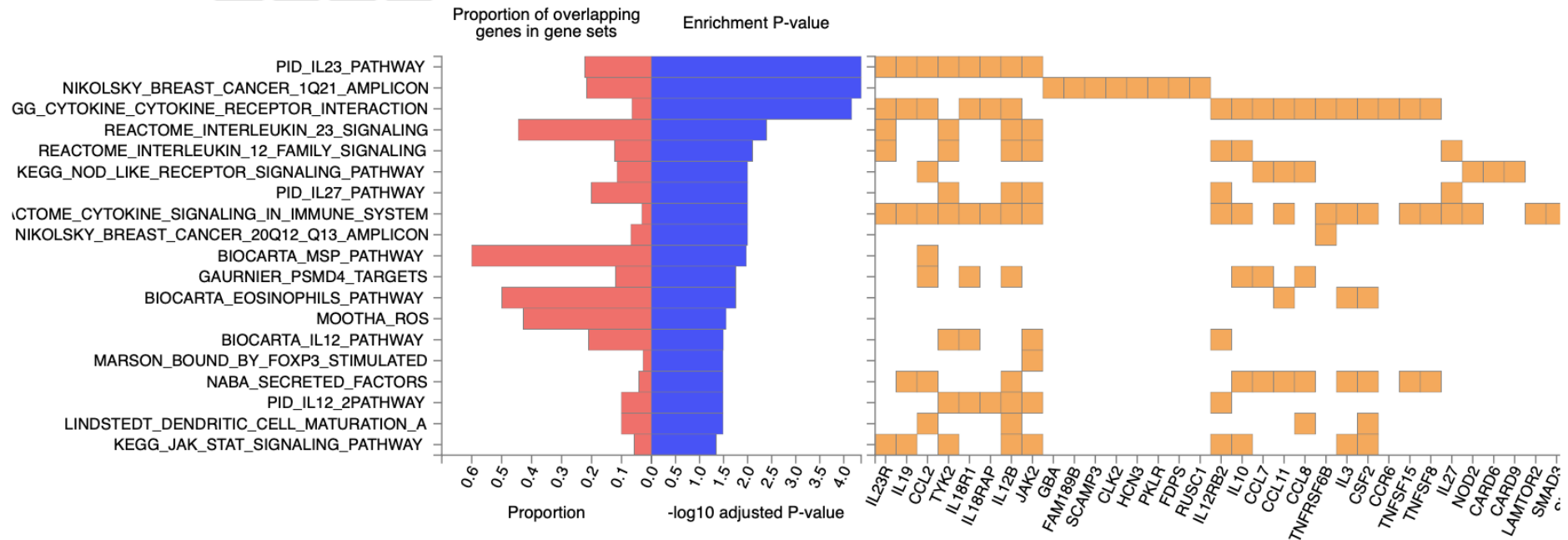
FUMA example

Are the genes enriched for certain pre-defined gene sets?

Curated_gene_sets (19)

[Plot / Table](#)

Download the plot as [PNG](#) [JPG](#) [SVG](#) [PDF](#)



FUMA example

Are the genes enriched for expression in certain tissue types?
(There are a lot of gene sets)



FUMA example

Annotation of the genes (links to gene/protein databases)

Links to external databases

Show **10** entries Search:

ENSG	entrezID	symbol	OMIM	UniProtID	DrugBank
ENSG00000001617	6405	SEMA3F	601124	Q13275	NA
ENSG00000003756	10181	RBM5	606884	P52756	NA
ENSG00000004534	10180	RBM6	606886	P78332	NA
ENSG00000008300	1951	CELSR3	604264	Q9NYQ7	NA
ENSG00000010256	7384	UQCRC1	191328	P31930	DB04141:DB04741:DB04799:DB07401:DB07763:DB07778:DB08330:DB0
ENSG00000026036	100533107	RTEL1- TNFRSF6B	NA	NA	NA
ENSG00000026297	8635	RNASET2	612944	O00584	NA
ENSG00000033627	535	ATP6V0A1	192130	Q93050	NA
ENSG00000049245	9341	VAMP3	603657	Q15836	NA
ENSG00000049246	8863	PER3	603427	P56645	NA

Showing 1 to 10 of 256 entries

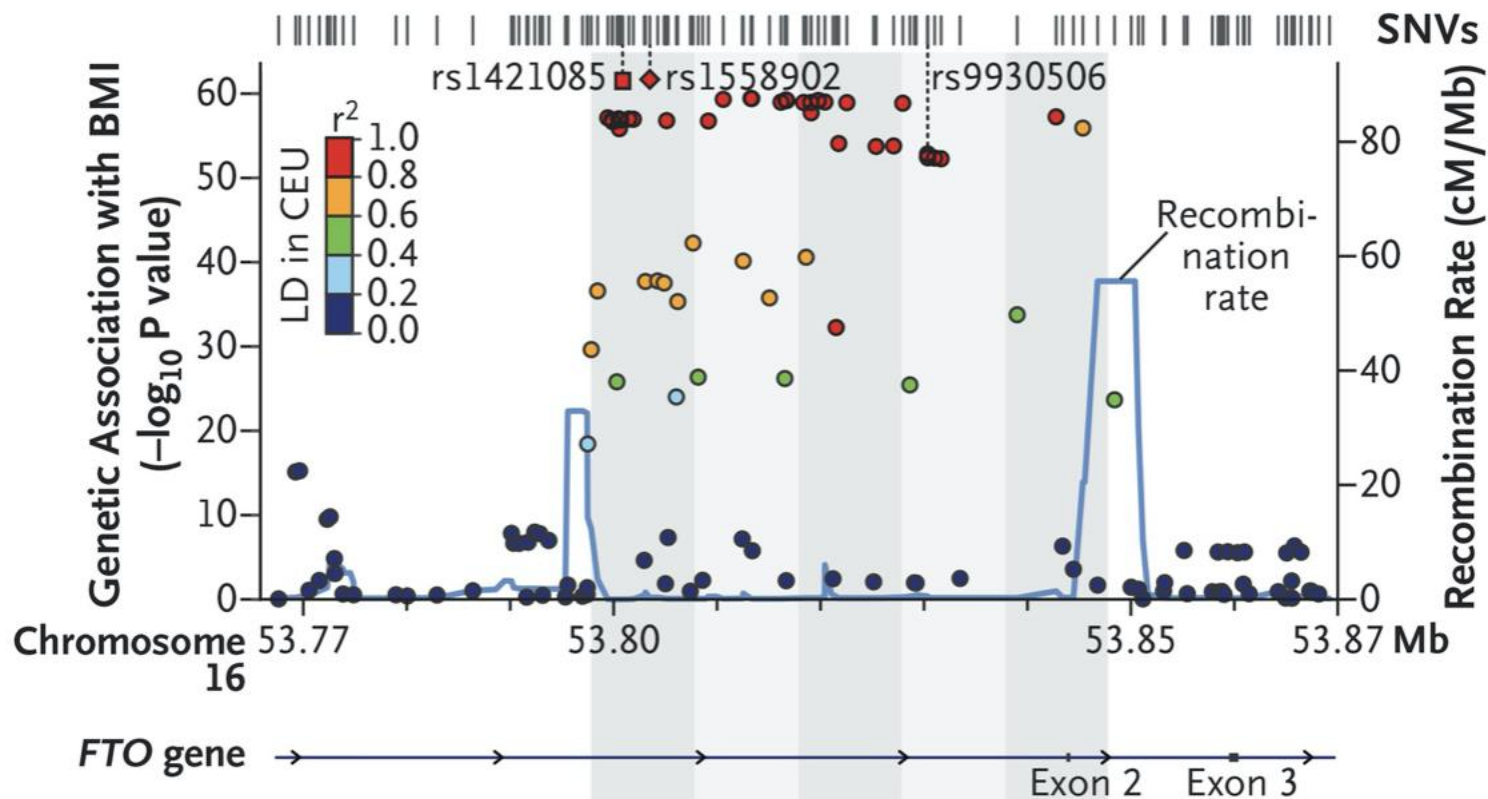
Previous 1 2 3 4 5 ...

26 Next

Annotation success story: *FTO* locus in BMI

- Strongest BMI association
- Intronic
- 89 SNPs over 47 kb
- Biological significance/mechanism of association was unclear (causal variant, target gene, cell type, pathway, mechanism all unknown)

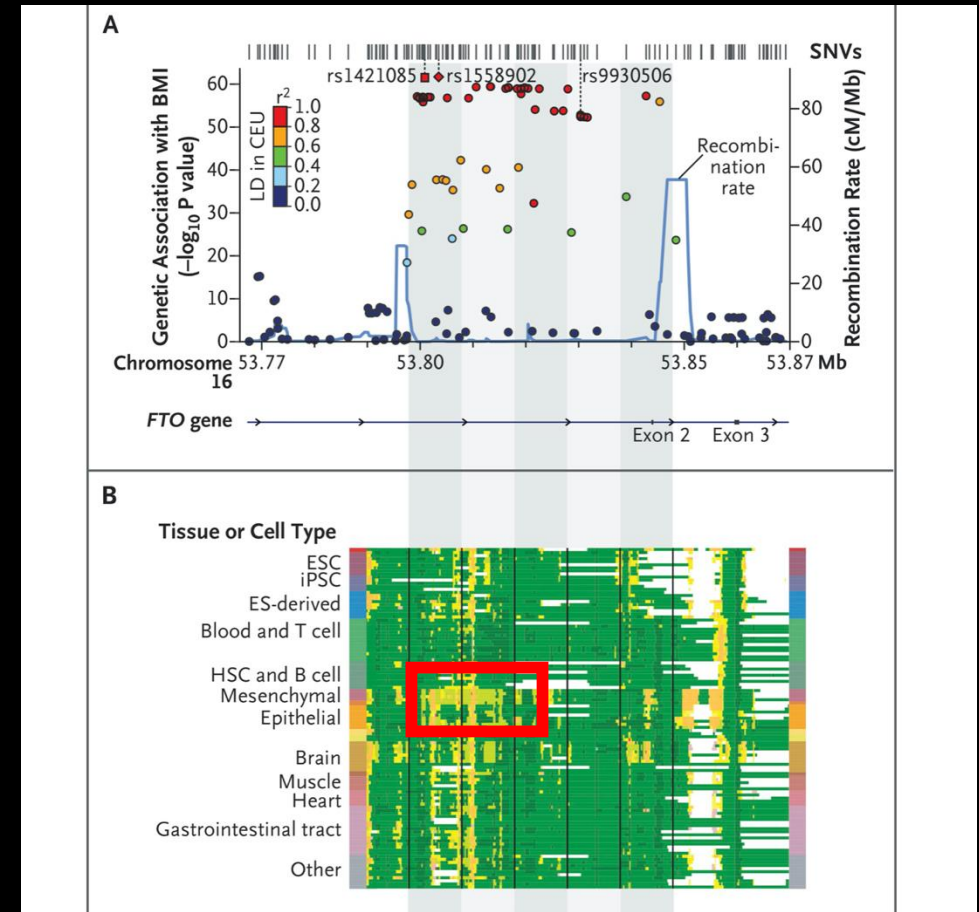
A



Claussnitzer, M., Dankel, S. N., Kim, K. H., Quon, G., Meuleman, W., Haugen, C., Glunk, V., Sousa, I. S., Beaudry, J. L., Puvion, V., Abdennur, N. A., Liu, J., Svensson, P. A., Hsu, Y. H., Drucker, D. J., Mellgren, G., Hui, C. C., Hauner, H., & Kellis, M. (2015). *FTO* Obesity Variant Circuitry and Adipocyte Browning in Humans. *The New England journal of medicine*, 373(10), 895–907. <https://doi.org/10.1056/NEJMoa1502214>

FTO annotation

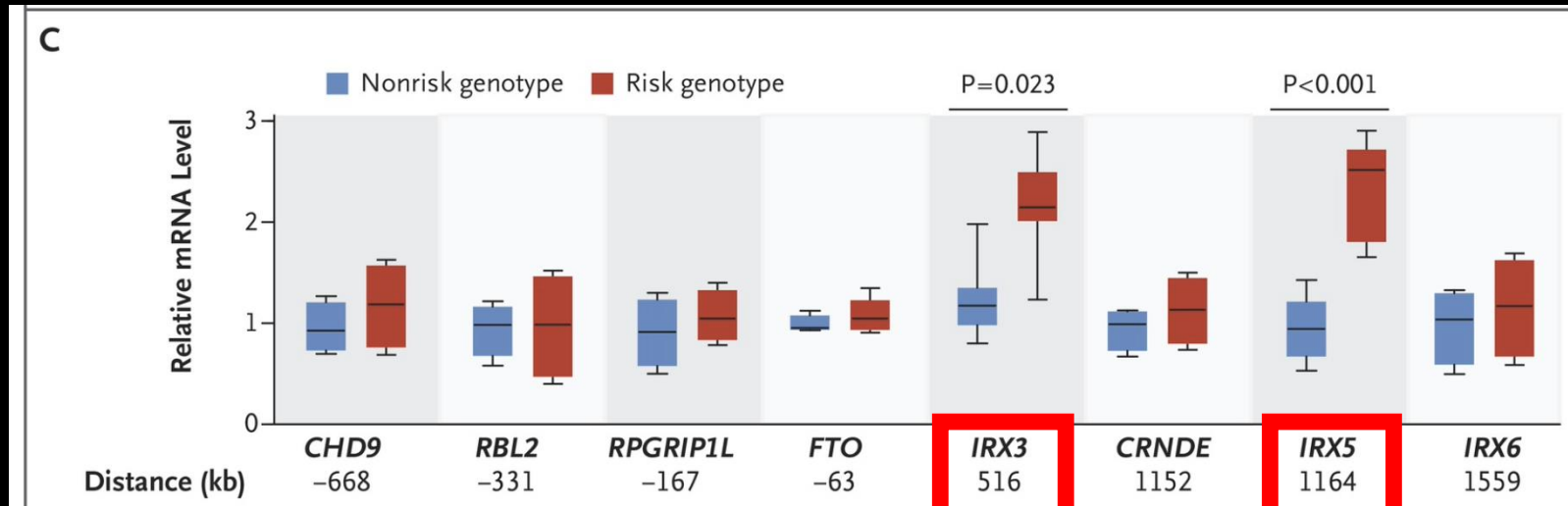
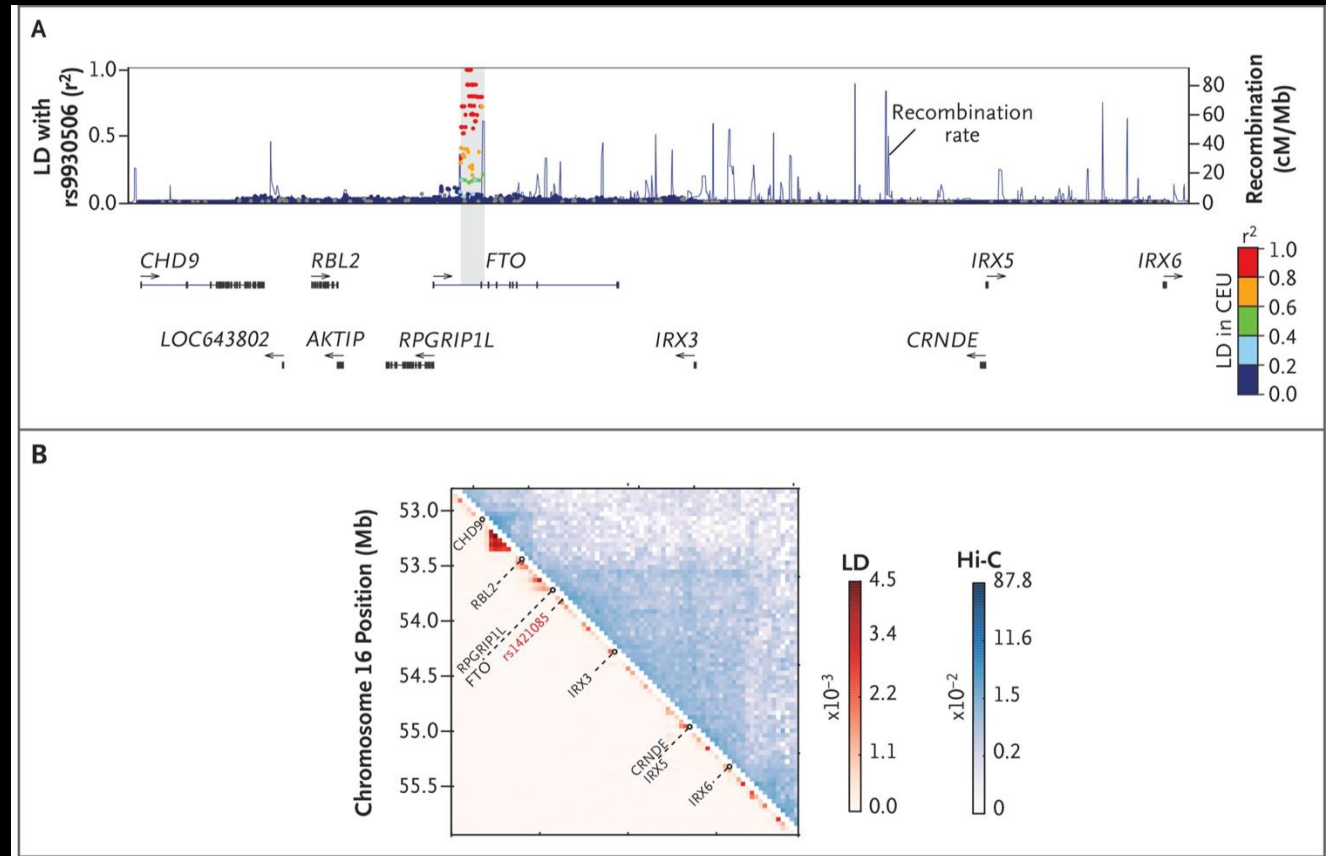
- Look at chromatin state across various cell types
- Noticed an “unusually long enhancer (12.8 kb) in mesenchymal adipocyte progenitor indicat[ing] a major regulatory locus ”
- (Some earlier “behavioral” hypotheses expected brain tissues to be “the” relevant tissues)



FTO annotation

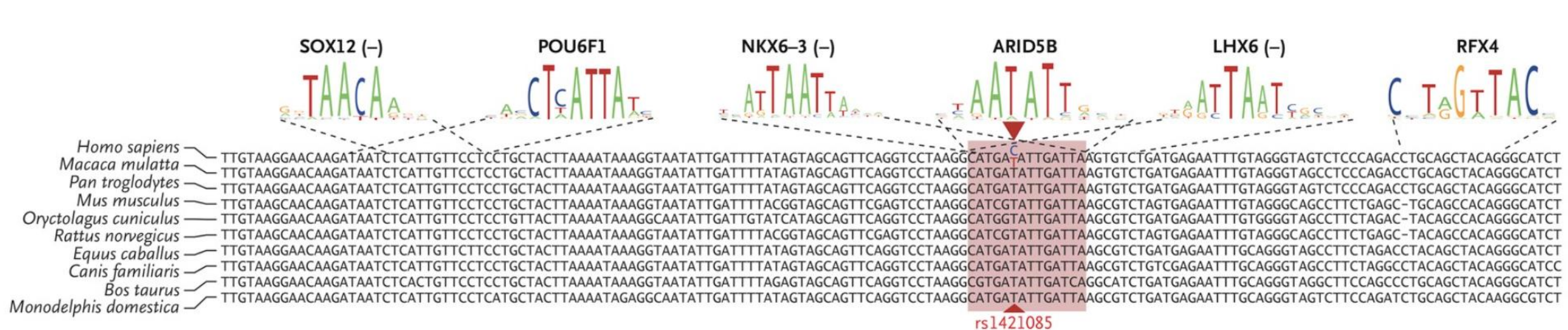
- A ~2 Mb “topological domain” includes 8 genes
- (What is a “topological domain”?)

- For these 8 genes, is expression associated with genotype at the top SNP?
- Yes, for *IRX3* and *IRX5*
- Risk allele → higher expression
- Effect limited to preadipocytes
- Not only cell *type* but also *stage* matters



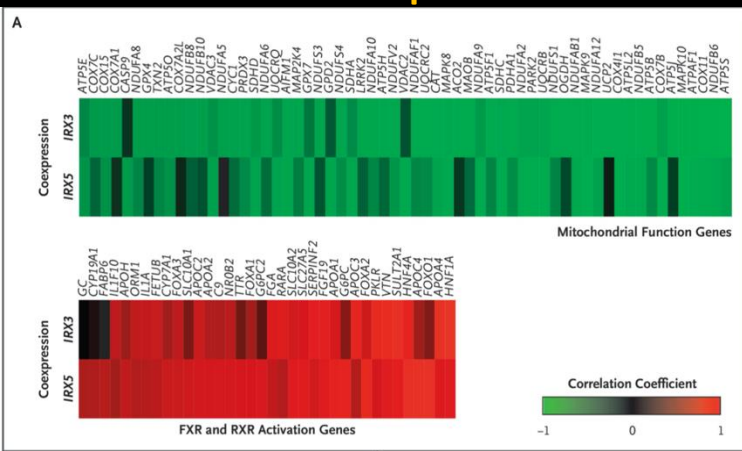
FTO annotation

- Why/how does the variant affect *IRX3/IRX5* expression?
- By altering “T” to “C” in a conserved *ARID5B* binding motif
- *ARID5B* binding represses transcription of *IRX3/IRX5*
- Disruption of binding → higher expression → phenotype
- (Details validated functionally)



FTO annotation

- Annotation success
 - Enhancer information → discover tissue (pre-adipocytes)
 - Topological information → candidate genes (including *IRX3/IRX5*)
 - Motif/conservation information → upstream regulator (*ARID5B*)
 - Causal SNP → gene expression → phenotype
- What else can we infer about the genes *IRX3/IRX5*?
 - What genes are co-expressed with them?
 - Implicates networks involving energy consumption



Another GWAS story

- Many LDL loci are in/near obvious candidates (like Mendelian dyslipidemia genes or established cholesterol target genes)
- 1p13 locus associated with myocardial infarction and LDL levels
- The association was strongest for this locus ($p=1E-170$)
- Same locus linked to coronary artery disease in other studies
- Homozygotes for risk (vs protective) alleles have ~40% higher MI risk and 16 higher mg/dl^{-1} LDL
- → Locus probably affects intermediate phenotype and clinical endpoint
- Mechanism unknown
 - Causal variant at 1p13?
 - Gene regulated by it?
 - Regulatory mechanism?
 - Mechanism of gene action on trait?

SORT1 (continued)

- Authors noted that the locus is associated with a specific class of LDL
- Strongest associations are non-coding, between *CELSR2* and *PRSC1* (functions unknown)
- Authors tested whether 1p13 SNPs are cis-regulators of nearby gene expression in liver and other “candidate tissues”
- 6 strongest SNPs form two haplotypes in people of European descent
- A minor allele of rs646776 affects expression of *CELSR2*, *PSRC1* and *SORT1* (the latter most); this was replicated

SORT1 (continued)

How to further fine-map?

The candidate region is defined by 6 SNPs in strong LD:

“Together these six best SNPs cluster in a noncoding DNA region that is 6.1 kilobases in size, spanning the 3' untranslated region (3'UTR) of *CELSR2*, the intergenic region, and the *PSRC1* 3'UTR oriented in the opposite direction”

“The six SNPs are in high linkage disequilibrium (LD) and comprise two predominant haplotypes in HapMap Europeans (CEU), with the ‘major’ haplotype present on 68% of chromosomes 1 and the ‘minor’ haplotype on 29%”

SORT1 (continued)

Test which SNP in the block appears to drive gene expression.

rs12740374 seems to be the important variant in the haplotype.

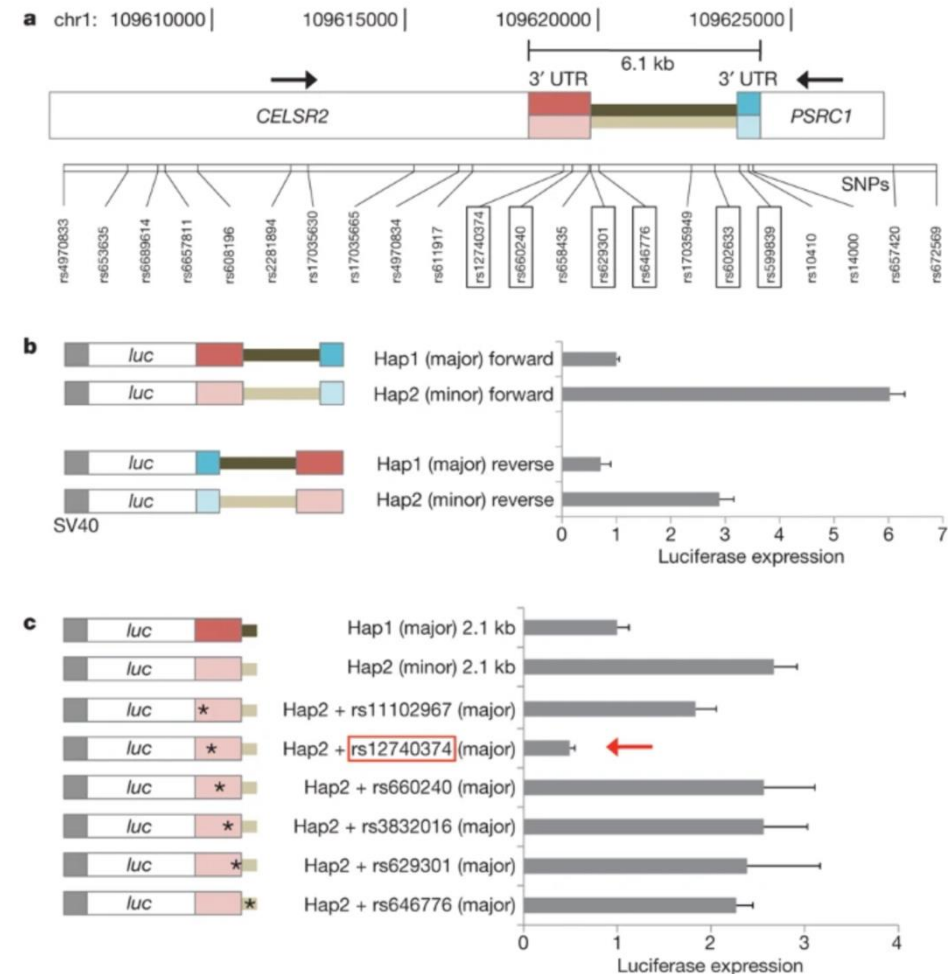
Specific to liver (not the two adipose tissue types).

How to further shrink the region?

Use LD from other populations.

“We genotyped rs12740374 and 15 other SNPs in or near the 6.1 kb noncoding region in ~9,000 African Americans. Whereas six SNPs have indistinguishable evidence for association with LDL-C in Europeans, we found that, in African Americans, rs12740374 alone had the strongest evidence for association ($P = 2.3 \times 10^{-20}$ for rs12740374 versus 9.2×10^{-15} at the next best 1p13 SNP)”

Figure 2: rs12740374 is responsible for haplotype-specific difference in transcriptional activity.



SORT1 (continued)

Further use of bioinformatics to determine SNP mechanism:

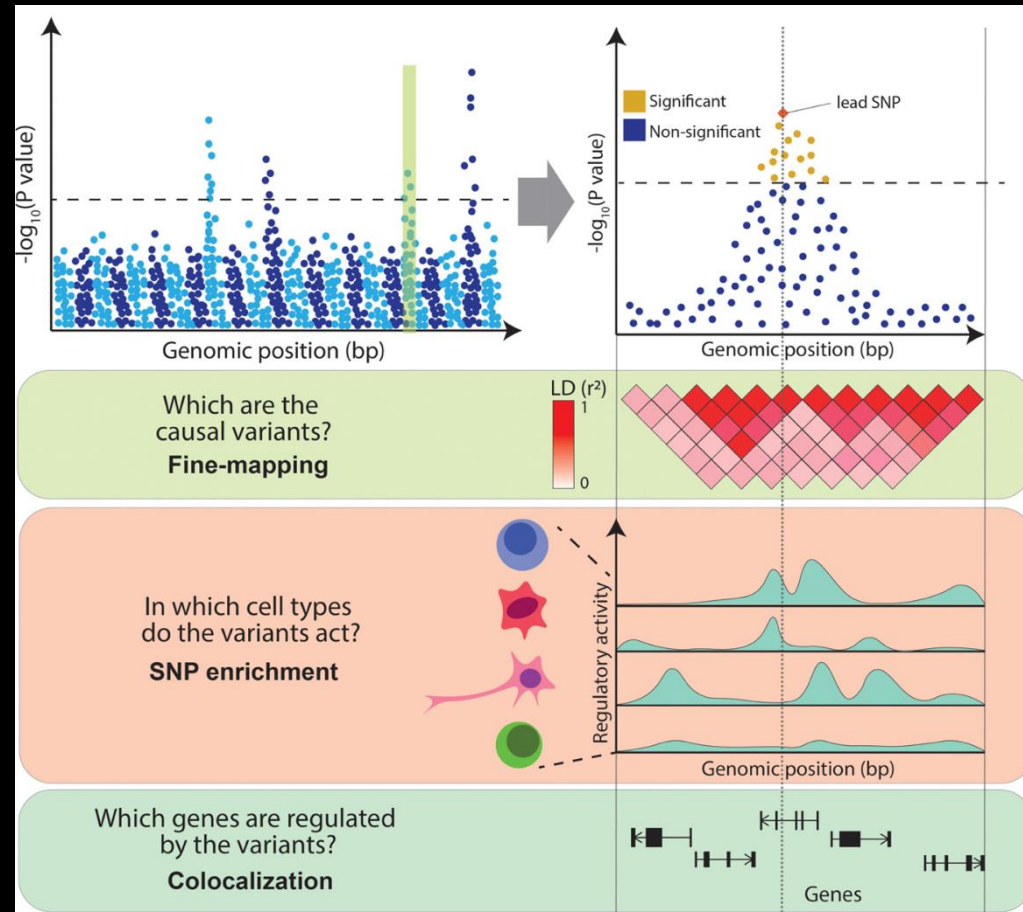
“We observed that rs12740374 alters a predicted binding site for C/EBP transcription factors, with the minor allele creating the site and the major allele disrupting it; the binding site is not present in the orthologous DNA region in mice (Fig. 3a). C/EBP α (also known as CEBPA) is a liver-enriched transcriptional factor that regulates the expression of numerous hepatic genes involved in a variety of metabolic processes”

Further functional validation was done.

Confirmed an rs12740374 allele-specific, tissue-specific change in binding.

Further confirmed that this affects *SORT1* expression the most.

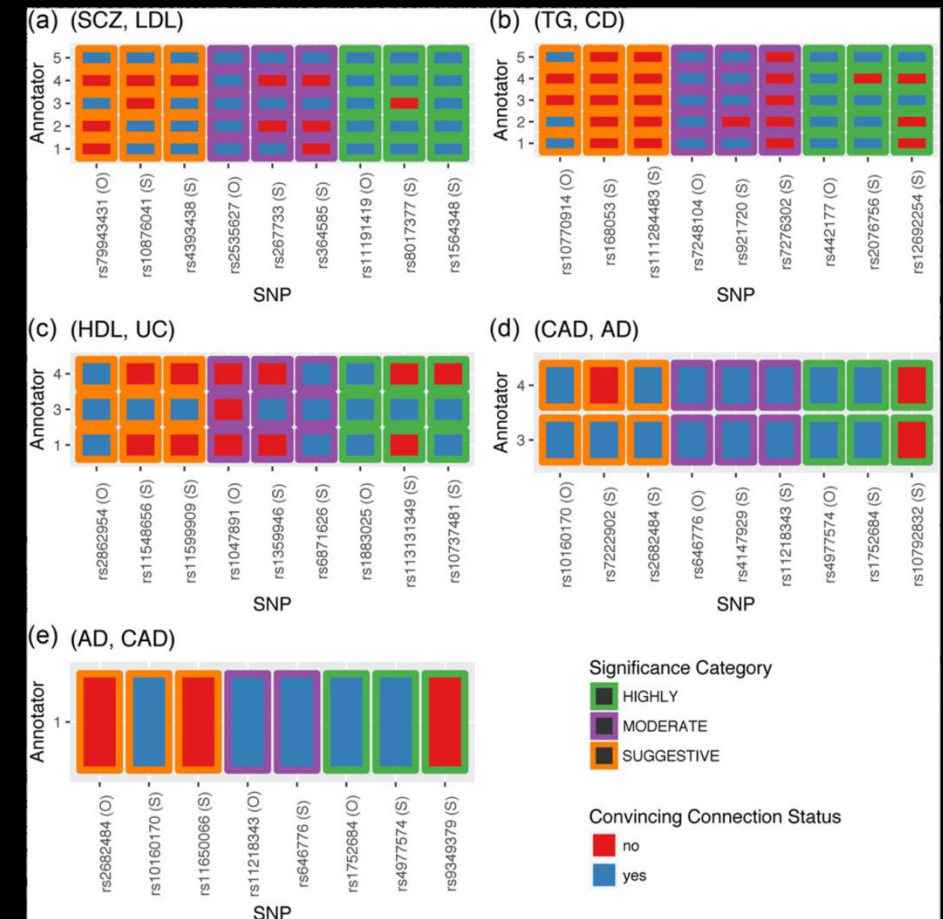
Summary



“Risks” of annotation

Abstract

When interpreting genome-wide association peaks, it is common to annotate each peak by searching for genes with plausible relationships to the trait. However, “all that glitters is not gold”—one might interpret apparent patterns in the data as plausible even when the peak is a false positive. Accordingly, we sought to see how human annotators interpreted association results containing a mixture of peaks from both the original trait and a genetically uncorrelated “synthetic” trait. Two of us prepared a mix of original and synthetic peaks of three significance categories from five different scans along with relevant literature search results and then we all annotated these regions. Three annotators also scored the strength of evidence connecting each peak to the scanned trait and the likelihood of further studying that region. While annotators found original peaks to have stronger evidence ($p_{\text{Bonferroni}} = 0.017$) and higher likelihood of further study ($p_{\text{Bonferroni}} = 0.006$) than synthetic peaks, annotators often made convincing connections between the synthetic peaks and the original trait, finding these connections 55% of the time. These results show that it is not difficult for annotators to make convincing connections between synthetic association signals and genes found in those regions.



Some risks and limitations

- Fooling oneself with a “nice story”
- Multiple testing problems (e.g., testing hundreds of tissues/cell types)
- Failure to “find something” doesn’t prove any null hypothesis about a particular variant or locus (e.g., maybe no one has studied the right gene/tissue/cell at the right time/under the right conditions)